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Yamashita et al.

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(45) **Date of Patent: *Sep. 9, 2025**

(54) **COMPUTING SYSTEM CONFIGURING DESTINATION ACCELERATORS BASED ON USAGE PATTERNS OF USERS OF A TRANSPORT SERVICE**

(71) Applicant: **Uber Technologies, Inc.**, San Francisco, CA (US)

(72) Inventors: **Yuhki Yamashita**, San Francisco, CA (US); **Didier Patrick Hilhorst**, San Francisco, CA (US); **Bryant Jow**, San Francisco, CA (US); **Peter Ng**, San Francisco, CA (US)

(73) Assignee: **Uber Technologies, Inc.**, San Francisco, CA (US)

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(63) Continuation of application No. 18/627,194, filed on Apr. 4, 2024, which is a continuation of application

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G06Q 50/40 (2024.01)
G06F 3/04842 (2022.01)

(Continued)

(52) **U.S. Cl.**

CPC ***G06Q 50/40*** (2024.01); ***G06F 3/04842*** (2013.01); ***G06F 3/0485*** (2013.01);

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(58) **Field of Classification Search**

CPC G06F 9/451; G06F 3/04842; G06F 3/0485; H04L 67/52; H04L 67/60; H04L 67/1095; H04L 67/306

See application file for complete search history.

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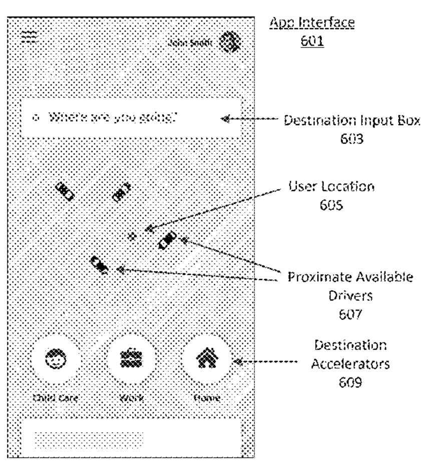
Primary Examiner — Rupangini Singh
(74) *Attorney, Agent, or Firm* — Mahamedi IP Law LLP

(57) **ABSTRACT**

A computing system can detect the launch of a rider application on computing devices of users of a transport service. The computing system can receive location data indicating the current location of each user, and determine a usage pattern for each user based on historical data corresponding to historical utilization of the transport service by the user. Based on the current location and the usage pattern of the user, the computing system can determine one or more suggested destination locations for the user, and transmit, over the one or more networks, display data to cause the rider application to display a destination accelerator for each of the one or more suggested destination locations. The destination accelerator can be selectable by the user to automatically input a destination location into a transport request for the transport service.

20 Claims, 10 Drawing Sheets



Home Screen 600

Related U.S. Application Data

No. 16/440,567, filed on Jun. 13, 2019, now Pat. No. 11,954,754, which is a continuation of application No. 15/395,341, filed on Dec. 30, 2016, now Pat. No. 10,417,727.

(60) Provisional application No. 62/399,642, filed on Sep. 26, 2016.

(51) **Int. Cl.**

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G06Q 10/1093	(2023.01)
H04L 67/1095	(2022.01)
H04L 67/306	(2022.01)
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H04L 67/60	(2022.01)

(52) **U.S. Cl.**

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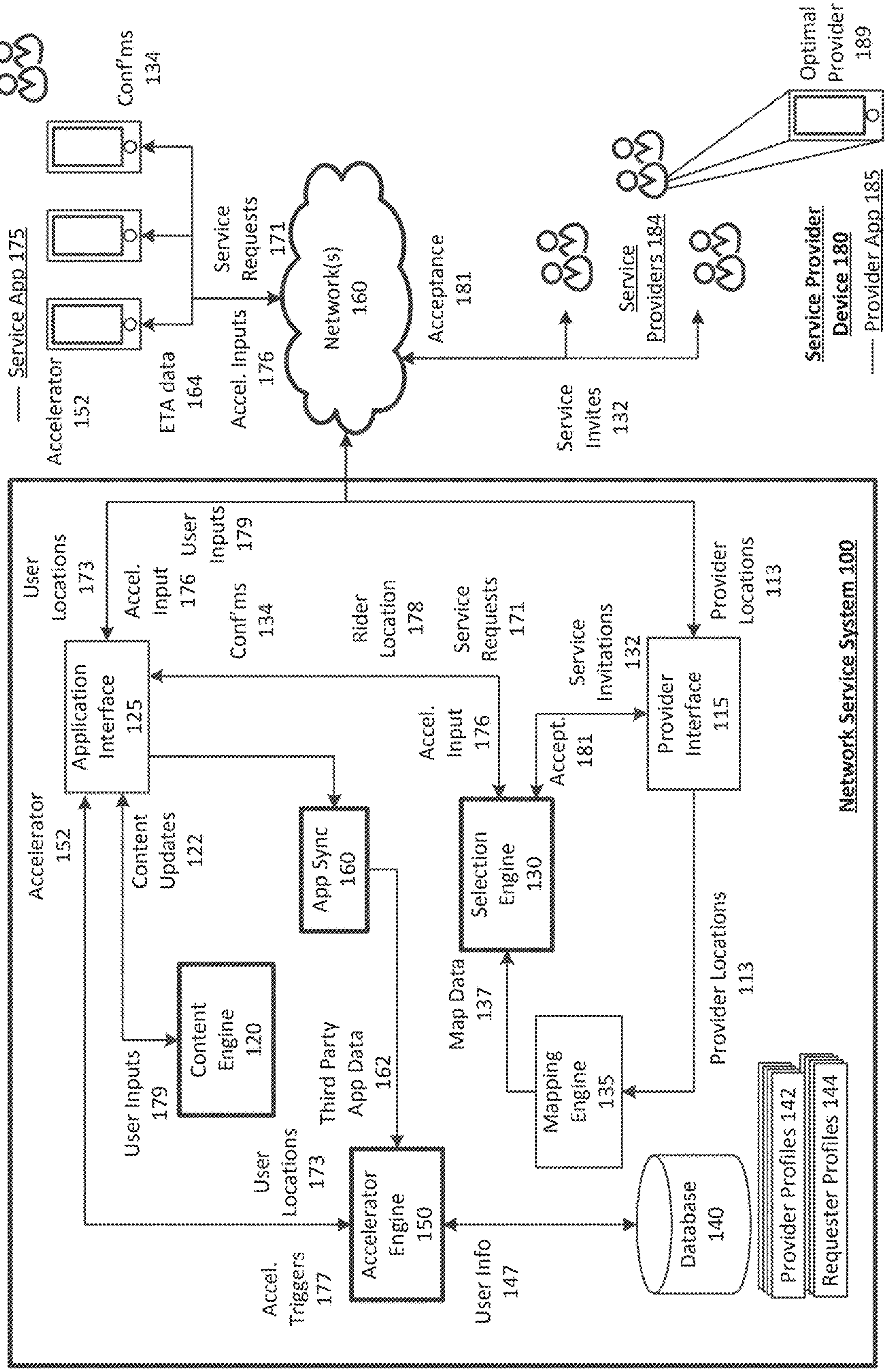
Requesting Users 174
Requester Devices 170
Service App 175
Accelerator 152
Conf'ms 134
ETA data 164
Accel. Inputs 176
Service Requests 171
Network(s) 160
Acceptance 181
Service Invites 132
Service Providers 184
Optimal Provider 189
Service Provider Device 180
Provider App 185
Network Service System 100
User Locations 173
Accel. Input 176
User Inputs 179
Conf'ms 134
Rider Location 178
Service Requests 171
Service Invitations 132
Provider Locations 113
Application Interface 125
Provider Interface 115
Accel. Input 176
Accept. 181
Accelerator 152
Content Updates 122
App Sync 160
Selection Engine 130
Provider Locations 113
Map Data 137
Content Engine 120
User Inputs 179
Third Party App Data 162
Mapping Engine 135
User Locations 173
Accel. Triggers 177
Accelerator Engine 150
User Info 147
Database 140
Provider Profiles 142
Requester Profiles 144

FIG. 1

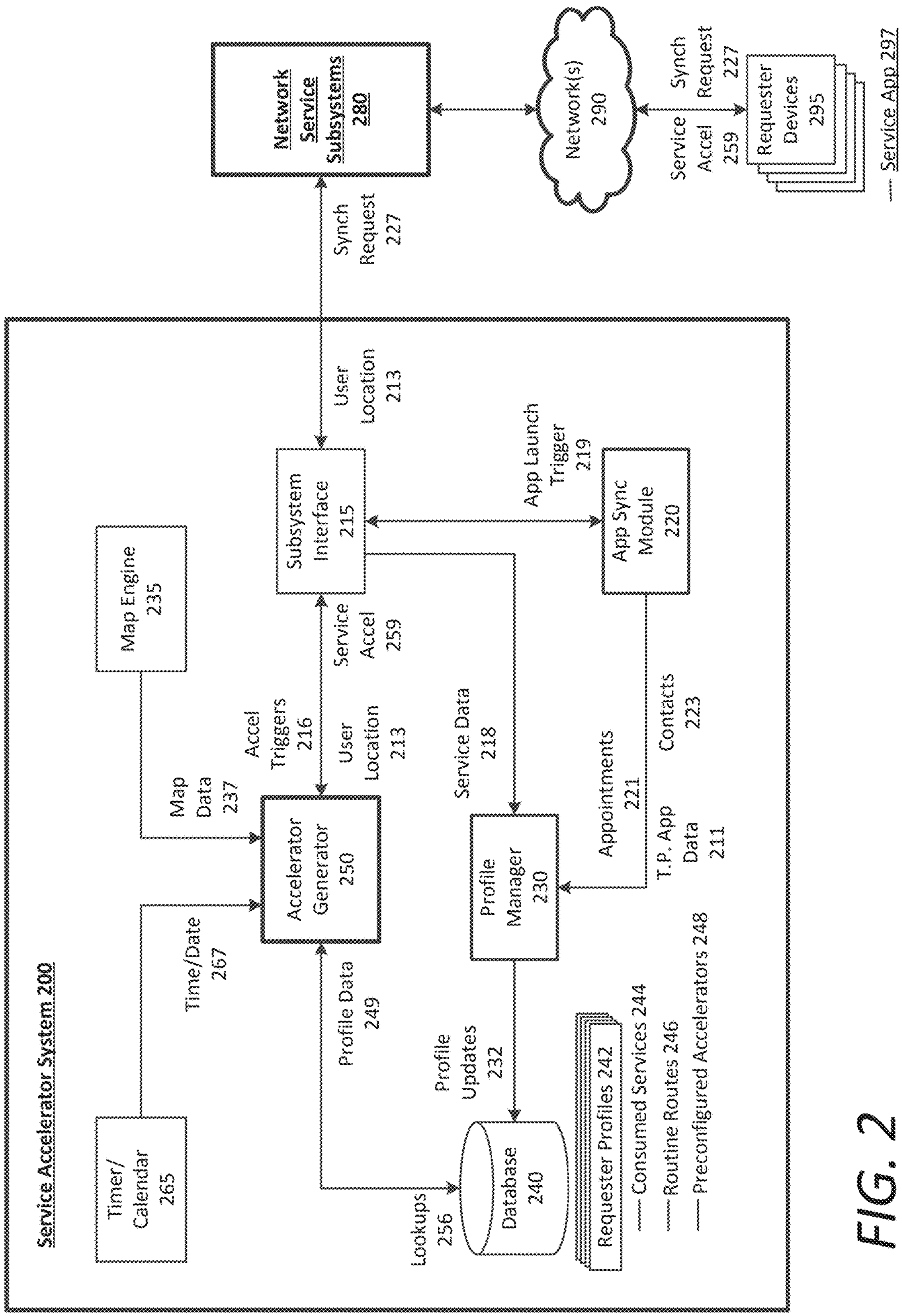
Service Accelerator System 200
Timer/ Calendar 265
Time/Date 267
Map Engine 235
Map Data 237
Accelerator Generator 250
Accel Triggers 216
User Location 213
Service Accel 259
Subsystem Interface 215
User Location 213
Synch Request 227
Network Service Subsystems 280
Network(s) 290
Service Accel 259
Synch Request 227
Requester Devices 295
Service App 297
App Launch Trigger 219
App Sync Module 220
Service Data 218
Appointments 221
Contacts 223
T.P. App Data 211
Profile Manager 230
Profile Data 249
Profile Updates 232
Lookups 256
Database 240
Requester Profiles 242
Consumed Services 244
Routine Routes 246
Preconfigured Accelerators 248

FIG. 2

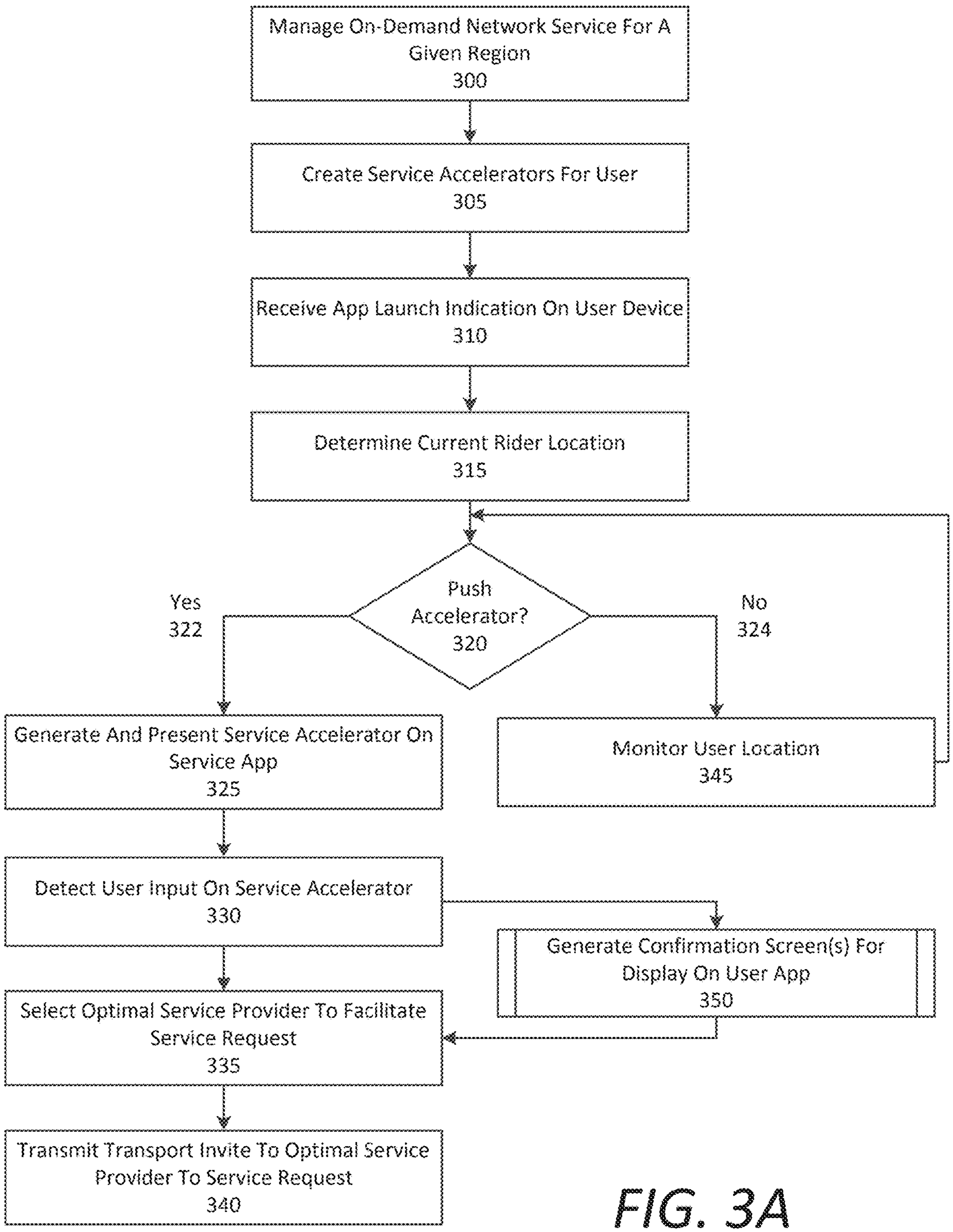
Manage On-Demand Network Service For A Given Region
300
Create Service Accelerators For User
305
Receive App Launch Indication On User Device
310
Determine Current Rider Location
315
Push Accelerator?
320
Yes
322
No
324
Generate And Present Service Accelerator On Service App
325
Monitor User Location
345
Detect User Input On Service Accelerator
330
Generate Confirmation Screen(s) For Display On User App
350
Select Optimal Service Provider To Facilitate Service Request
335
Transmit Transport Invite To Optimal Service Provider To Service Request
340

FIG. 3A

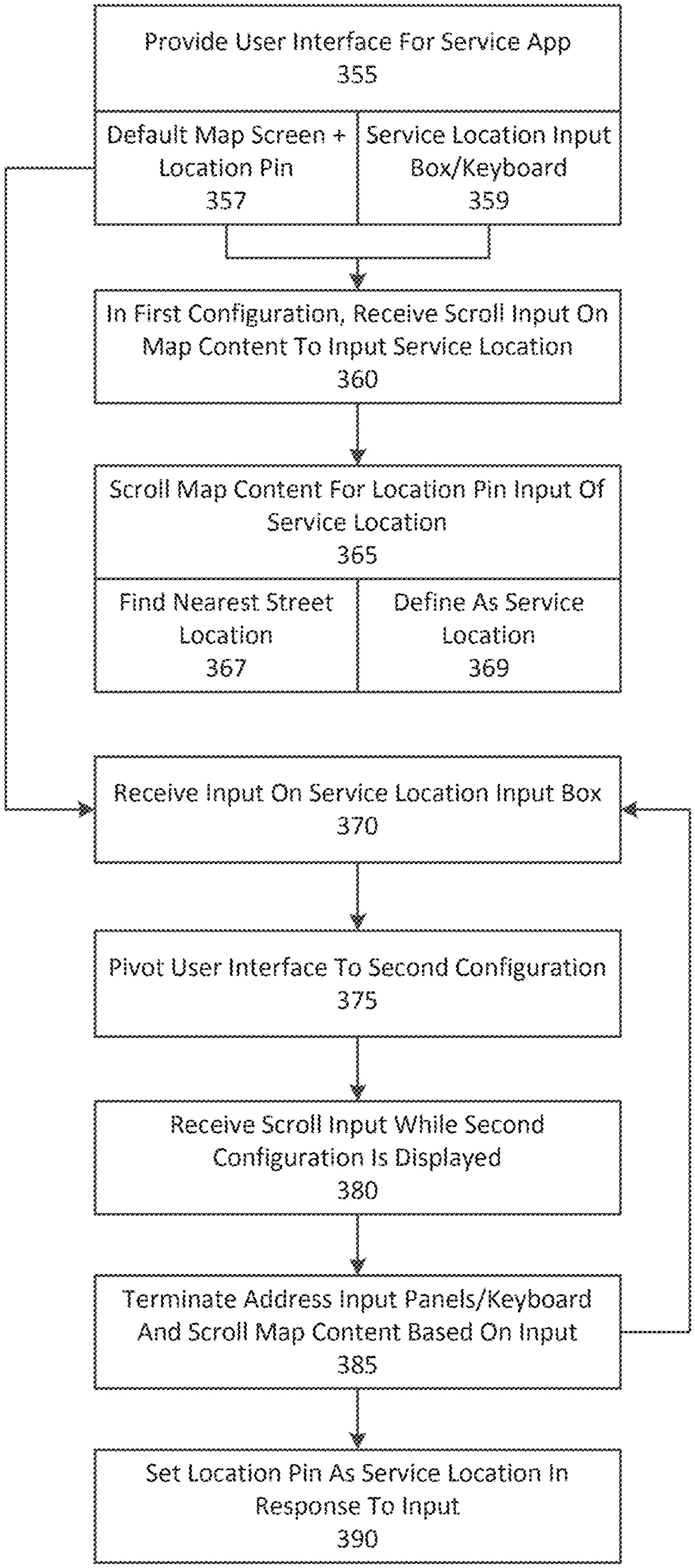
Provide User Interface For Service App
355
Default Map Screen + Location Pin
357
Service Location Input Box/Keyboard
359
In First Configuration, Receive Scroll Input On Map Content To Input Service Location
360
Scroll Map Content For Location Pin Input Of Service Location
365
Find Nearest Street Location
367
Define As Service Location
369
Receive Input On Service Location Input Box
370
Pivot User Interface To Second Configuration
375
Receive Scroll Input While Second Configuration Is Displayed
380
Terminate Address Input Panels/Keyboard And Scroll Map Content Based On Input
385
Set Location Pin As Service Location In Response To Input
390

FIG. 3B

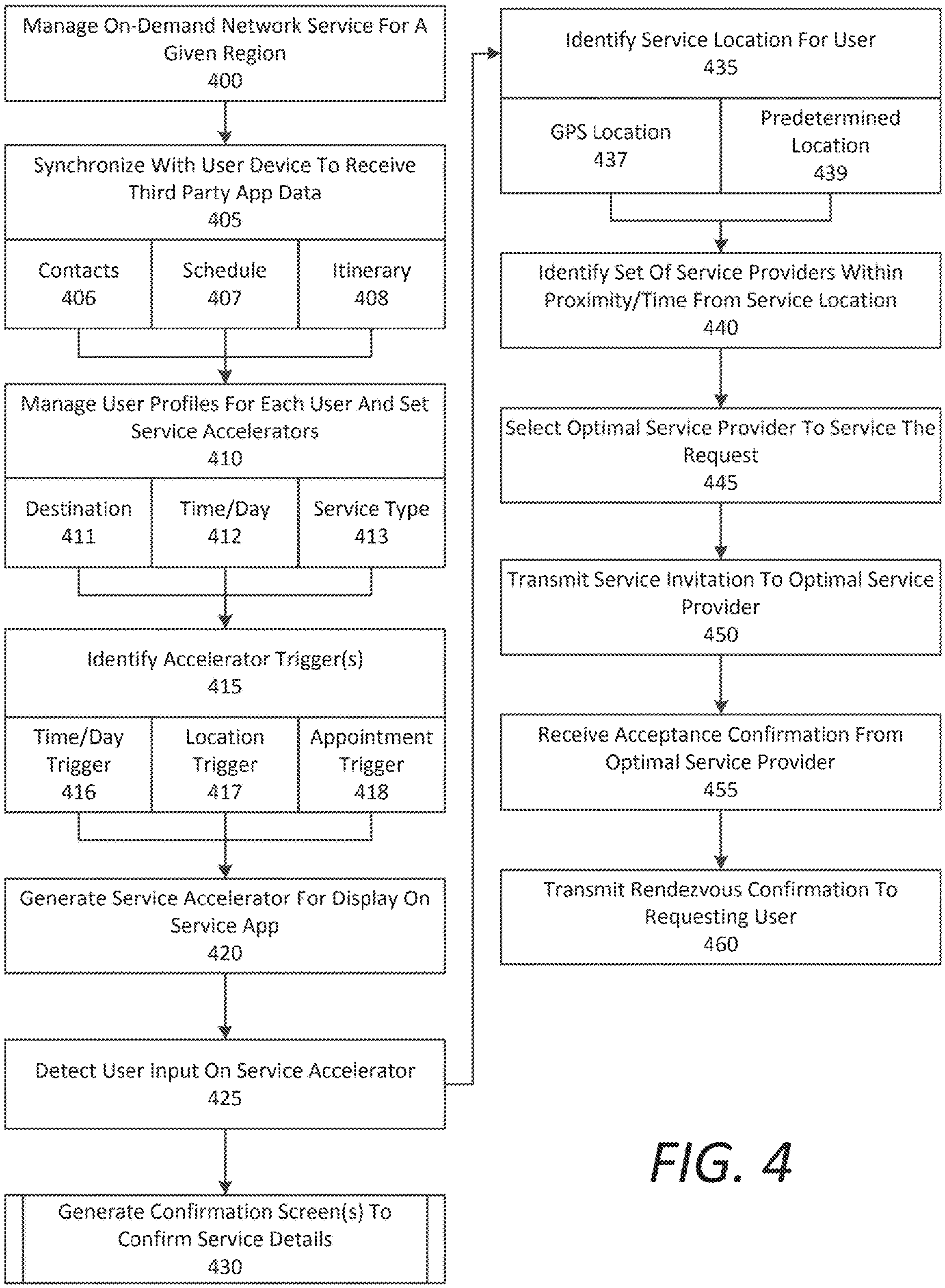
Manage On-Demand Network Service For A Given Region
400
Synchronize With User Device To Receive Third Party App Data
405
Contacts
406
Schedule
407
Itinerary
408
Manage User Profiles For Each User And Set Service Accelerators
410
Destination
411
Time/Day
412
Service Type
413
Identify Accelerator Trigger(s)
415
Time/Day Trigger
416
Location Trigger
417
Appointment Trigger
418
Generate Service Accelerator For Display On Service App
420
Detect User Input On Service Accelerator
425
Generate Confirmation Screen(s) To Confirm Service Details
430
Identify Service Location For User
435
GPS Location
437
Predetermined Location
439
Identify Set Of Service Providers Within Proximity/Time From Service Location
440
Select Optimal Service Provider To Service The Request
445
Transmit Service Invitation To Optimal Service Provider
450
Receive Acceptance Confirmation From Optimal Service Provider
455
Transmit Rendezvous Confirmation To Requesting User
460

FIG. 4

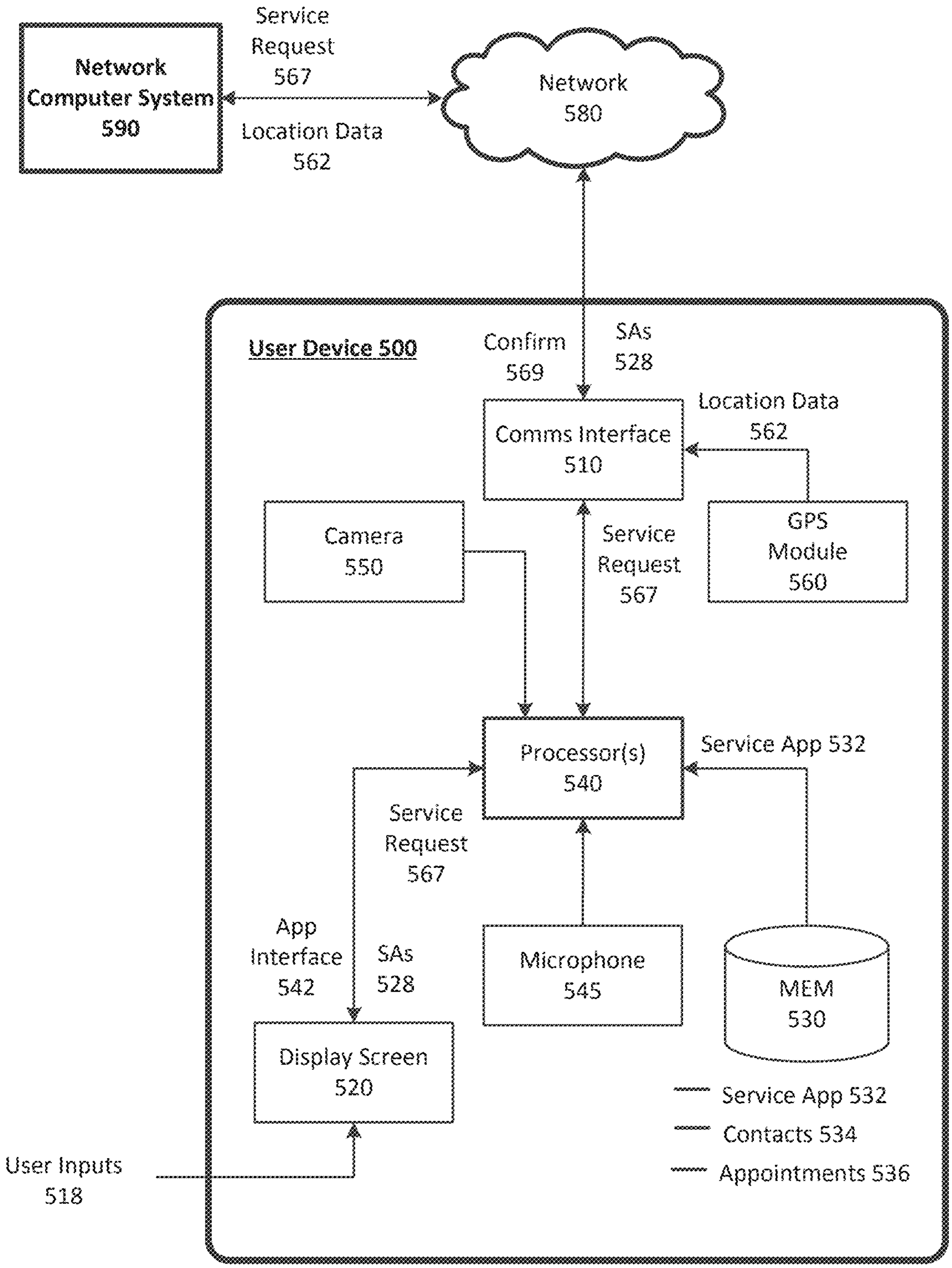
Network Computer System 590
Service Request 567
Location Data 562
Network 580
User Device 500
Confirm 569
SAs 528
Comms Interface 510
Location Data 562
Camera 550
Service Request 567
GPS Module 560
Processor(s) 540
Service App 532
Service Request 567
App Interface 542
SAs 528
Microphone 545
MEM 530
Display Screen 520
Service App 532
Contacts 534
Appointments 536
User Inputs 518

FIG. 5

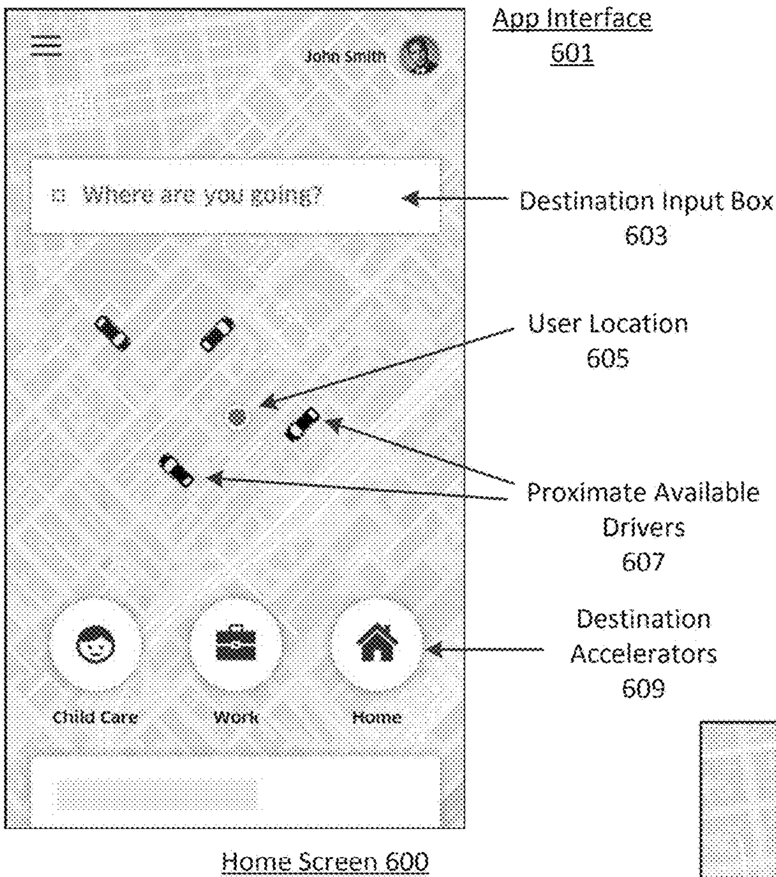
App Interface
601
John Smith
Where are you going?
Destination Input Box
603
User Location
605
Proximate Available
Drivers
607
Destination
Accelerators
609
Child Care
Work
Home
Home Screen 600

FIG. 6A

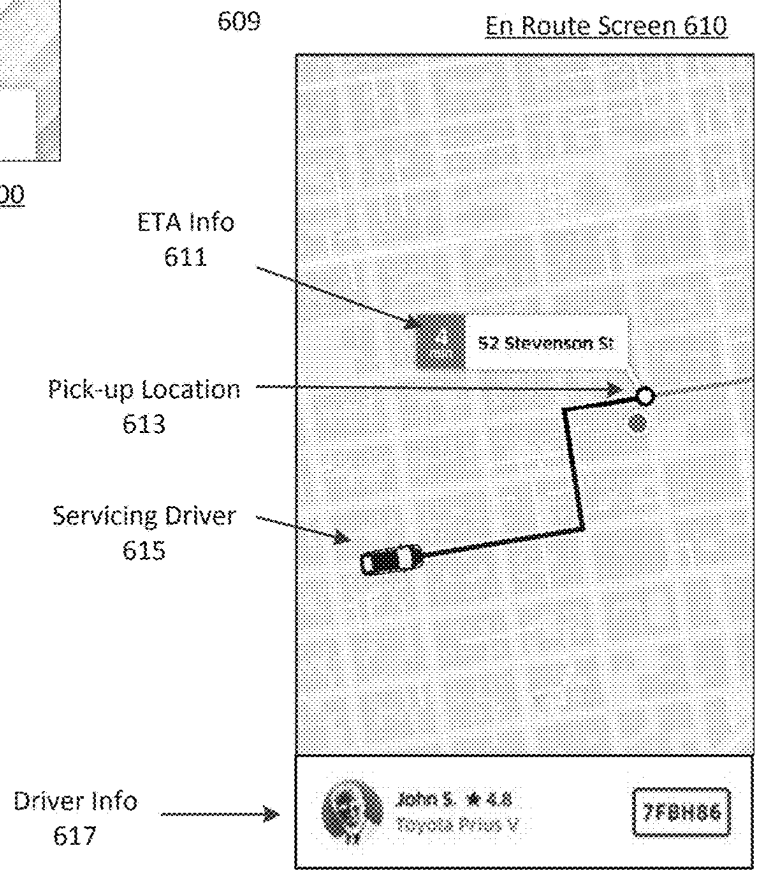
En Route Screen 610
ETA Info
611
52 Stevenson St
Pick-up Location
613
Servicing Driver
615
Driver Info
617
John S. ★ 4.8
Toyota Prius V
7FBH86

FIG. 6B

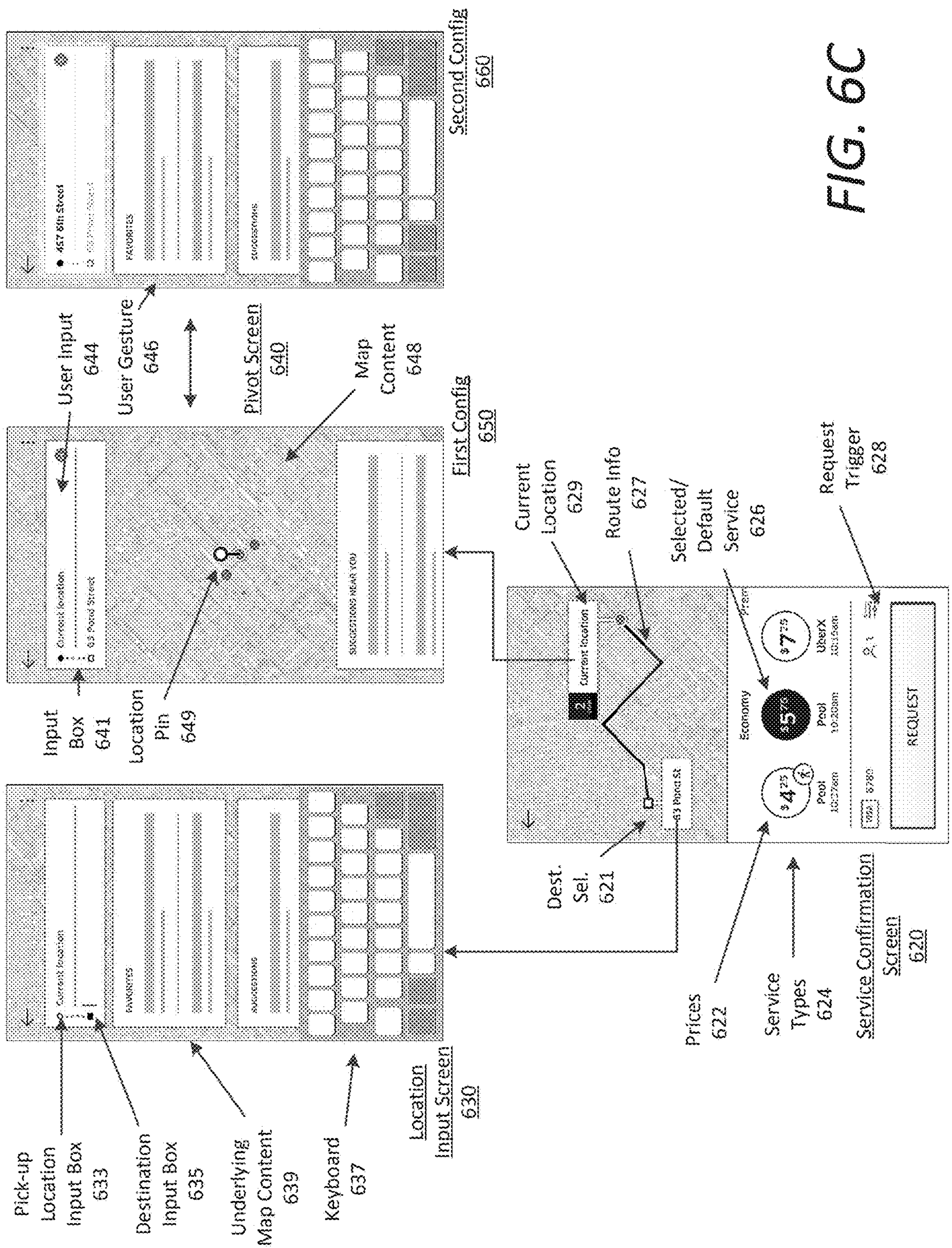
Pick-up Location Input Box 633
Destination Input Box 635
Underlying Map Content 639
Keyboard 637
Location Input Screen 630
Input Box 641
Location Pin 649
User Input 644
User Gesture 646
Pivot Screen 640
Map Content 648
First Config 650
Second Config 660
Current Location 629
Route Info 627
Selected/ Default Service 626
Request Trigger 628
Dest. Sel. 621
Prices 622
Service Types 624
Service Confirmation Screen 620
Current location
63 Pond Street
457 6th Street
FAVORITES
SUGGESTIONS
SUGGESTIONS NEAR YOU
63 Pond St
Economy
Pool
UberX
REQUEST

FIG. 6C

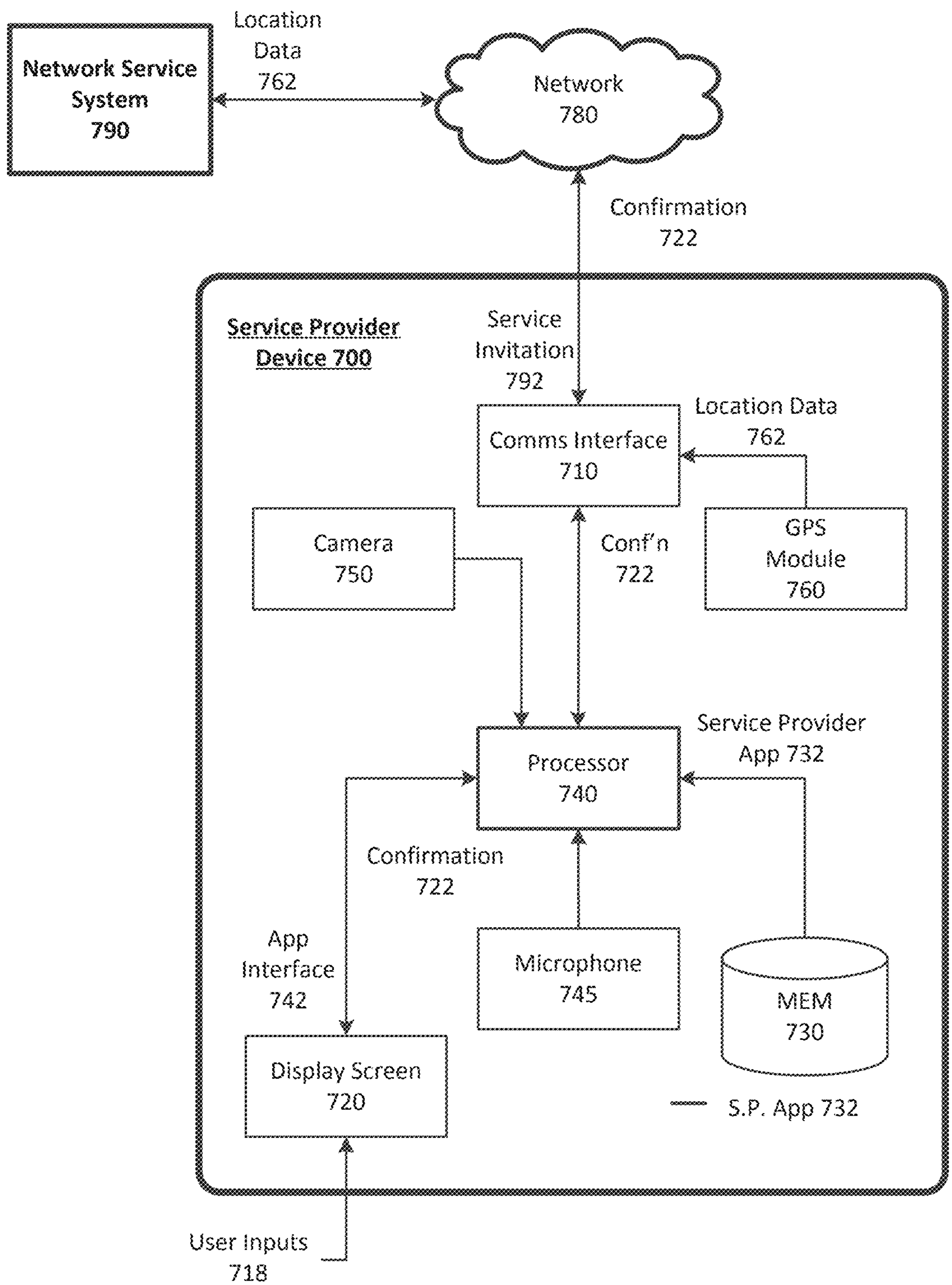
Network Service System 790
Location Data 762
Network 780
Confirmation 722
Service Provider Device 700
Service Invitation 792
Comms Interface 710
Location Data 762
Camera 750
Conf'n 722
GPS Module 760
Service Provider App 732
Processor 740
Confirmation 722
App Interface 742
Microphone 745
MEM 730
Display Screen 720
S.P. App 732
User Inputs 718

FIG. 7

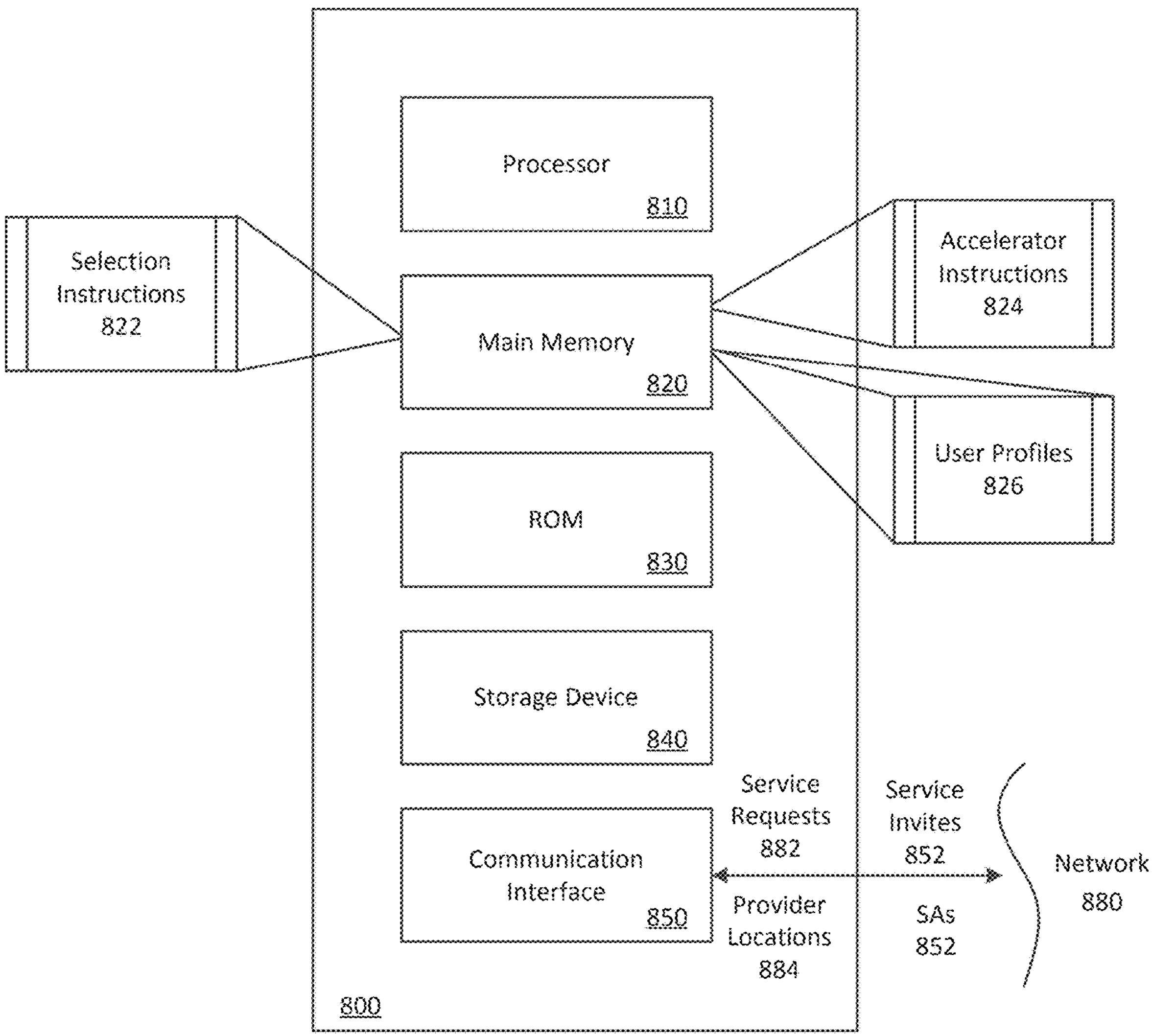
Processor
810
Selection Instructions
822
Main Memory
820
Accelerator Instructions
824
User Profiles
826
ROM
830
Storage Device
840
Communication Interface
850
Service Requests
882
Provider Locations
884
Service Invites
852
SAs
852
Network
880
800

FIG. 8

COMPUTING SYSTEM CONFIGURING DESTINATION ACCELERATORS BASED ON USAGE PATTERNS OF USERS OF A TRANSPORT SERVICE

CROSS REFERENCE TO RELATED APPLICATION

This application is a continuation of U.S. patent application Ser. No. 18/627,194, filed on Apr. 4, 2024; which is a continuation of U.S. patent application Ser. No. 16/440,567, filed on Jun. 13, 2019, now U.S. Pat. No. 11,954,754 issued Apr. 9, 2024; which is a continuation of U.S. patent application Ser. No. 15/395,341, filed on Dec. 30, 2016; now U.S. Pat. No. 10,417,727, issued Sep. 17, 2019; which claims the benefit of priority to U.S. Provisional Application No. 62/399,642, filed on Sep. 26, 2016; the aforementioned applications being hereby incorporated by reference in their respective entireties.

BACKGROUND

User centric network services typically sequence users through a number of selection interfaces so that the user can specify certain information for a desired type of service, including service level selections and preferences With enhancements in network and mobile technology, the number of on-demand services for user selection is also increasing, creating inconvenience for human operators. Moreover, the time needed for selection can occupy an interface device, creating performance issues and draining resources of the operative selection device.

BRIEF DESCRIPTION OF THE DRAWINGS

The disclosure herein is illustrated by way of example, and not by way of limitation, in the figures of the accompanying drawings in which like reference numerals refer to similar elements, and in which:

FIG. **1** is a block diagram illustrating an example network computer system in communication with user and service provider devices, in accordance with examples described herein;

FIG. **2** is a block diagram illustrating an example destination accelerator system utilized in connection with a network computer system, according to examples described herein;

FIG. **3**A is a flow chart describing an example method of generating a service accelerator, according to examples described herein;

FIG. **3**B is a flow chart describing an example method of pivoting a user interface to enable input of one or more service locations;

FIG. **4** is a flow chart describing an example method of configuring a service accelerator and facilitating on-demand services, according to examples described herein;

FIG. **5** is a block diagram illustrating an example user device executing a designated service application for an on-demand network-based service, as described herein;

FIGS. **6**A through **6**C illustrate example screenshots of a user interface on a user device, according to examples described herein;

FIG. **7** is a block diagram illustrating an example service provider device executing a designated service provider application to facilitate an on-demand network-based service, as described herein; and

FIG. **8** is a block diagram illustrating a computer system upon which examples described herein may be implemented.

DETAILED DESCRIPTION

A network computer system is provided herein that manages an on-demand network-based service linking available service providers with service requesters throughout a given region (e.g., a metroplex such as the San Francisco Bay Area). In doing so, the network computer system can receive service requests for on-demand services from requesting users via a designated service application executing on the users' mobile computing devices. Based on a detected service location or an inputted service location (e.g., a pick-up location), the network computer system can identify a number of proximate available service providers (e.g., available drivers) and transmit a service invitation to one or more service provider devices of the proximate available service providers to fulfill the service request. In many examples, the service providers can either accept or decline the service invitation based on, for example, the service location being impractical for the service provider.

In determining an optimal service provider to fulfill or complete a given service request, the network computer system can identify a plurality of candidate service providers to fulfill the service request based on a service location indicated in the service request. For example, the network computer system can determine a geo-fence surrounding the service location (or a geo-fence defined by a radius away from the service location), identify a set of candidate service providers (e.g., twenty or thirty service providers within the geo-fence), and select an optimal service provider (e.g., a closest service provider to the service location, a service provider with the shortest estimated travel time to the service location, a service provider traveling to a location within a specified distance or specified travel time to the service location, etc.) from the candidate service providers to fulfill the service request. According to examples provided herein, the network computer system can compile historical data for individual users with regard to the on-demand network-based service. Thus, the network computer system can manage a user profile for each user indicating routine start and/or end locations (or regions), and/or routine routes (e.g., for a transportation service from home to work and/or vice versa) and/or preferred service types (e.g., transportation, delivery, mailing, etc.). In some examples, the network computer system can further synchronize with a user device to, for example, identify the user's contacts, the user's schedule and appointments, travel plans (e.g., a scheduled trip), and the like.

In various implementations, the network computer system can establish a set of criteria that determines when a service accelerator is to be displayed on the user device. According to examples herein, a service accelerator refers to a selectable feature on a client application that, when selected by a user, automatically associates a transport or service request with a set of parameters, preconfigured settings based on contextual information, and/or a set of service criteria. In some examples, the network computer system can associate a particular location (e.g., a work or home location) with a time of day (or a time period of day), and establish a service accelerator for the user based on the location and time of day. For example, the historical data for the user may indicate that the user routinely utilizes the transport service (e.g., twice a week or multiple times a month, etc.) to travel home on weekdays at or around 7:30 pm. Thus, the network computer system can generate a "home" service accelerator that can be displayed on the user's device (e.g., a user interface corresponding to the executing service application on the user device) a certain time period prior to 7:30 pm on every weekday (e.g., 5:00 pm-7:30 pm or even to a time after 7:30 pm, alternatively, such as 5-9 pm). As an addition or an alternative, the network computer system can use contextual information associated with the user, such as where the user currently is, to determine whether or not to display the "home" service accelerator (e.g., if the user is currently at or within a predefined distance of the user's home or is located in another state or country, the "home" service accelerator would not be displayed). As another example, historical data for the user may indicate that the user routinely utilizes the on-demand service to go to the gym after work (e.g., around 6 pm or 7 pm multiple times a week). Thus, the network computer system can generate a "gym" service accelerator for the user for display on the user's device a certain time prior to the end of the workday.

According to examples described herein, selection of a service accelerator by the user can automatically cause the network computer system to receive, from the user device, a service request, and process the service request without any additional input or subsequent manual intervention by the user (e.g., begin selection process to select a proximate service provider to service the ride). In some examples, the service accelerator, or accelerator feature, can greatly reduce the number of sequential screens needed by current transport service applications in order to confirm a service request. In other examples, selection of a service accelerator can cause a confirmation screen to be displayed (e.g., confirming the estimated service cost, service type, pick-up location and/or destination, service time, etc.) on the client application. In such examples, the service accelerator can save the user the trouble of having to manually input certain items such as a pick-up location and/or destination, and/or manually selecting a service type, and the user can select a request or confirm feature from the confirmation screen without having to manually input other information in order to make a request for the service (e.g., a two-click request after opening or launching the client application).

In variations, a single selection of the displayed service accelerator can automatically cause the network computer system to configure the on-demand service with preconfigured settings for the service type, and utilize the user's current location or address to determine an ideal service location. Additionally or alternatively, the service accelerator can be displayed with an estimated or upfront price of the service (e.g., a computed value before a service is requested), the service type, and/or an estimated time of arrival of a nearest service provider. In some aspects, the upfront price or estimated price can be displayed with individual service accelerators based on current price and service type that corresponds to the service accelerator (e.g., when there is high GPS confidence of the current location of the user). Accordingly, the user may be presented with important data corresponding to the resultant ride along with the service accelerator itself. Furthermore, upon confirmation of the service request, the user's device can display information indicating that the service provider is en route to rendezvous with the user (e.g., traveling to the pick-up location), and information to enable the user to identify the service provider and/or the service provider's vehicle when the service provider approaches the rendezvous location.

In various examples, multiple service accelerators may be displayed on the user interface of the service application at any given time. Some service accelerators may be persistent, or displayed by default—such as those associated with the user's routine destinations. For example, whenever the user is away from home, the network computer system can generate a home service accelerator (i.e., a home "destination" accelerator) for display on the user's device to enable the user to readily request a ride home. In variations, the network computer system can provide the home service accelerator at certain times of the day and/or days of the week, or when the user's current location is a predefined distance away from the user's home location (e.g., two miles) and/or within a predefined distance of the user's home location (e.g., fifty miles). As an example, the network computer system can push home service accelerators to user devices on Friday and/or Saturday nights to aid in the overall prevention of driving under the influence (e.g., conditioned upon the user being away from home or at a restaurant, bar, or nightclub location).

Additionally, the network computer system can analyze the historical data associated with the user's profile to determine any particular patterns in the user's routines with regard to the on-demand network-based service. For example, the user may routinely request service at a particular location on a certain day of the month or year. In variations, the network computer system can determine a pattern in which the user meets another user of the on-demand service in a routine manner, and can correlate the two users to generate a service accelerator that specifies a person as opposed to a specified location. As such, the service accelerator can be selected by the user to request transportation to the other user's current location, or to a scheduled meeting location for both users. Along these lines, the network computer system can synchronize with other applications on the computing devices of the users, and can correlate schedules and/or appointments of an individual user, or between multiple users of the transport service. Thus, service accelerators can be generated in accordance with third party applications on the computing devices of users, and can also include other user locations in addition to specific destination locations.

As provided herein, the terms "user" and "service requester" may used throughout this application interchangeably to describe a person who utilizes a service application on a computing device to request, over one or more networks, on demand services from a network computing system. Furthermore, the terms "service accelerator" and "accelerator feature" may be used in this application interchangeably to represent a displayed feature or icon on the service requester's computing device that, when selected, provides an accelerated process of requesting the on-demand service. For example, selection of the "service accelerator" or "accelerator feature" can cause the computing device of the service requester to automatically determine a location where the on-demand service is to be fulfilled or completed, or where a service provider is to rendezvous with the service requester.

Among other benefits, the examples described herein achieve a technical effect of improving user experience with regard to utilizing an on-demand network-based service. The use of service accelerators leverages historical user data to provide the user with a more straightforward, time-sensitive, and intuitive approach to utilizing the on-demand network-based service.

As used herein, a computing device refers to devices corresponding to desktop computers, cellular devices or smartphones, personal digital assistants (PDAs), laptop computers, virtual reality (VR) or augmented reality (AR) headsets, tablet devices, television (IP Television), etc., that can provide network connectivity and processing resources for communicating with the system over a network. A computing device can also correspond to custom hardware, in-vehicle devices, or on-board computers, etc. The computing device can also operate a designated application configured to communicate with the network-based service.

One or more examples described herein provide that methods, techniques, and actions performed by a computing device are performed programmatically, or as a computer-implemented method. Programmatically, as used herein, means through the use of code or computer-executable instructions. These instructions can be stored in one or more memory resources of the computing device. A programmatically performed step may or may not be automatic.

One or more examples described herein can be implemented using programmatic modules, engines, or components. A programmatic module, engine, or component can include a program, a sub-routine, a portion of a program, or a software component or a hardware component capable of performing one or more stated tasks or functions. As used herein, a module or component can exist on a hardware component independently of other modules or components. Alternatively, a module or component can be a shared element or process of other modules, programs or machines.

Some examples described herein can generally require the use of computing devices, including processing and memory resources. For example, one or more examples described herein may be implemented, in whole or in part, on computing devices such as servers, desktop computers, cellular or smartphones, personal digital assistants (e.g., PDAS), laptop computers, VR or AR devices, printers, digital picture frames, network equipment (e.g., routers) and tablet devices. Memory, processing, and network resources may all be used in connection with the establishment, use, or performance of any example described herein (including with the performance of any method or with the implementation of any system).

Furthermore, one or more examples described herein may be implemented through the use of instructions that are executable by one or more processors. These instructions may be carried on a computer-readable medium. Machines shown or described with figures below provide examples of processing resources and computer-readable mediums on which instructions for implementing examples disclosed herein can be carried and/or executed. In particular, the numerous machines shown with examples of the invention include processors and various forms of memory for holding data and instructions. Examples of computer-readable mediums include permanent memory storage devices, such as hard drives on personal computers or servers. Other examples of computer storage mediums include portable storage units, such as CD or DVD units, flash memory (such as carried on smartphones, multifunctional devices or tablets), and magnetic memory. Computers, terminals, network enabled devices (e.g., mobile devices, such as cell phones) are all examples of machines and devices that utilize processors, memory, and instructions stored on computer-readable mediums. Additionally, examples may be implemented in the form of computer-programs, or a computer usable carrier medium capable of carrying such a program.

Some examples are referenced herein in context of an autonomous vehicle (AV) or self-driving vehicle (SDV). An AV or SDV refers to any vehicle which is operated in a state of automation with respect to steering and propulsion. Different levels of autonomy may exist with respect to AVs. For example, some vehicles may enable automation in limited scenarios, such as on highways, provided that drivers are present in the vehicle. More advanced AVs can drive without any human assistance from within or external to the vehicle. Such vehicles are often required to make advanced determinations regarding how the vehicle behaves given challenging surroundings of the vehicle environment.

System Descriptions

FIG. **1** is a block diagram illustrating an example network computer system in communication with user and service provider devices, in accordance with examples described herein. The network computer system **100** can manage an on-demand network service that connects requesting users **174** with service providers **184** that are available to service the users' **174** service requests **171**. The on-demand network-based service can provide a platform that enables on-demand services (e.g., ride sharing services) between requesting users **174** and available service providers **184** by way of a service application **175** executing on the requester devices **170**, and a service provider application **185** executing on the service provider devices **180**. As used herein, a requester device **170** and a service provider device **180** can comprise a computing device with functionality to execute a designated application corresponding to the transportation arrangement service managed by the network computer system **100**. In many examples, the requester device **170** and the service provider device **180** can comprise mobile computing devices, such as smartphones, tablet computers, VR or AR headsets, on-board computing systems of vehicles, and the like. Example on-demand network-based services can comprise on-demand delivery, package mailing, shopping, construction, plumbing, home repair, housing or apartment sharing, etc., or can include transportation arrangement services implementing a ride sharing platform.

The network computer system **100** can include an application interface **125** to communicate with requester devices **170** over one or more networks **160** via a service application **175**. According to examples, a requesting user **174** wishing to utilize the on-demand network-based service can launch the service application **175** and transmit a service request **171** over the network **160** to the network computer system **100**. In certain implementations, the requesting user **174** can view multiple different service types managed by the network computer system **100**, such as ride-pooling, a basic ride share services, a luxury vehicle service, a van or large vehicle service, a professional driver service (e.g., where the service provider is certified), a self-driving vehicle transport service, and the like. The network computer system **100** can utilize the service provider locations **113** to provide the requester devices **170** with ETA data **164** of proximate service providers for each respective service. For example, the service application **175** can enable the user **174** to scroll through each service type. In response to a soft selection of a particular service type, the network computer system **100** can provide ETA data **164** on a user interface of the service app **175** that indicates an ETA of the closest service provider for the service type, and/or the locations of all proximate available service providers for that service type. As the user scrolls through each service type, the user interface can update to show visual representations of service providers (e.g., vehicles) for that service type on a map centered around the user **174** or a pick-up location set by the user. The user can interact with the user interface of the service app **175** to select a particular service type, and transmit a service request **171**.

In some examples, the service request **171** can include a service location within a given region (e.g., a pick-up location within a metropolitan area managed by one or more datacenters corresponding to the network computer system **100**) in which a matched service provider is to rendezvous with the requesting user **174**. The service location can be inputted by the user by setting a location pin on a user interface of the service app **175**, or can be determined by a current location of the requesting user **174** (e.g., utilizing location-based resources of the requester device **170**). Additionally, for on-demand transport services, the requesting user **174** can further input a destination during or after submitting the service request **171**.

In various implementations, the network computer system **100** can further include a selection engine **130** to process the service requests **171** in order to ultimately select service providers **184** to fulfill the service requests **171**. The network computer system **100** can include a service provider interface **115** to communicate with the service provider devices **180** via the service provider application **185**. In accordance with various examples, the service provider devices **180** can transmit their current locations using location-based resources of the service provider devices **180** (e.g., GPS resources). These service provider locations **113** can be utilized by the selection engine **130** to identify a set of candidate service providers **184**, in relation to the service location, that are available to fulfill the service request **171**.

In certain implementations, the network computer system **100** can select a proximate self-driving vehicle (SDV) to service the service request **171** (e.g., a transportation request). Thus, the pool of proximate candidate service providers in relation to a rendezvous location can also include one or more SDVs operating throughout the given region.

In some aspects, the network computer system **100** can include a mapping engine **135**, or can utilize a third-party mapping service, to generate map data **137** and or traffic data in the environment surrounding the service location. The mapping engine **135** can receive the service provider locations **113** and input them onto the map data **137**. The selection engine **130** can utilize the current locations **113** of the service providers in the map data **137** (e.g., by setting a geo-fence surrounding the service location) in order to select an optimal service provider **189** to fulfill the service request **171**. As provided herein, the optimal service provider **189** can be a service provider that is closest to the requesting user **174** with respect to distance or time, or can be a proximate service provider that is optimal for other reasons, such as the service provider's experience, the amount of time the service provider has been on the clock, the service provider's current earnings, and the like.

Once the optimal service provider **189** is selected, the selection engine **130** can generate a service invitation **132** to service the service request **171**, and transmit the service invitation **132** to the optimal service provider's **189** device **180** via the service provider application **185**. Upon receiving the service invitation **132**, the optimal service provider **189** can either accept or reject the invitation **132**. Rejection of the invitation **132** can cause the selection engine **130** to determine another optimal service provider from the candidate set of service providers **184** to fulfill the service request **171**. However, if the optimal service provider accepts (e.g., via an acceptance input), then the acceptance input **181** can be transmitted back to the selection engine **130**, which can generate and transmit a confirmation **134** of the optimal service provider **189** to the requesting user **174** via the service application **175** on the requesting user's **174** computing device **170**.

According to examples provided herein, the network computer system **100** can include a content engine **120** that manages the manner in which content is displayed on the requester devices **170** and/or the service provider devices **180**. Regarding the requester devices **170**, the content engine **120** can provide content updates based on user inputs **179** on a user interface generated by the service application **175**. For example, a user selection on a content feature of the service app **175** can cause the content engine **120** to generate a new screen on the service app **175**, or cause a current screen to pivot between certain displayed features. When inputting a particular service location, the user may utilize a location pin and map content, and set the location pin on a particular location in the map content to input the service location. Additionally, the content engine **120** can cause a service location input box to overlay the map content, which can enable the requesting user **174** to select the input box to cause additional features to be displayed on the user interface (e.g., overlaying the map content). In variations, to return to the map content, the user **174** can input a gesture—such as a scroll or swipe gesture—anywhere on the screen. In response to the gesture, the content engine **120** can cause the additional features to dismiss, and re-enable map content scrolling with the location pin. These dynamically pivoting interfaces can be provided by the content engine **120** for the pick-up location input, the service location input, or both. Further description of the dynamically pivoting interfaces on the service application **175** is provided below with respect to FIG. **6**C.

In various implementations, the network computer system **100** can further include a database **140** storing requester profiles **144** including historical information specific to the individual users **174** of the on-demand network-based service. Such information can include user preferences of service types, routine services, service locations, pick-up and destination locations, work addresses, home addresses, addresses of frequently visited locations (e.g., a gym, grocery store, mall, local airport, sports arena or stadium, concert venue, local parks, and the like). In addition, the database **140** can further store service provider profiles **142** indicating information specific to individual service providers, such as vehicle type, service qualifications, earnings data, and service provider experience.

According to examples, the network computer system **100** can include an accelerator engine **150** which can provide individualized service accelerators **152** to the requester devices **170** based on accelerator triggers **179**. As provided herein, the accelerator triggers **179** can comprise current conditions that satisfy certain criteria for displaying a particular service accelerator **152** on a particular user's computing device **170**. As a basic example, a home destination accelerator for an on-demand transport service would not be practical if the user **174** was already located at the user's home. Thus, a criterion for the home destination accelerator could be that the current location of the user must be different from the user's home location. More intricate service accelerators **152** may be associated with multiple criteria, such as a time window, a service type, certain days of the week, a location or area, appointment or scheduling data, and the like. In certain variations, the user **174** can manually configure the set of criteria for a service accelerator **152** via a configuration interface or settings feature of the service application **175**. For example, the service application **175** can enable the user **174** configure a service accelerator **152** by selecting a service location and a service type so that a single input on the service accelerator **152** can cause the on-demand service to be arranged automatically, for the specified service type (e.g., a ride service to transport the user **174** to a specified destination). According to examples provided herein, when the accelerator triggers **179** satisfy a set of criteria for a specified service accelerator **152**, the accelerator engine **150** can generate the specified service accelerator **152**, and cause the accelerator **152** to be displayed on the requester device **170**.

In configuring the set of criteria for each service accelerator **152**, the accelerator engine **150** can analyze the user information **147** in the requester profiles **144**. In some examples, certain default service accelerators **152** can be stored for the individual user, such as a home destination accelerator, a work destination accelerator, a gym destination accelerator, or an accelerator for a destination frequently visited by the user **174**. In such examples, the service accelerator **152** for each of these default destinations may be displayed on the service app **175** (e.g., a home screen of the service app **175** for an on-demand transportation service) whenever the user **174** is not located at or within a certain proximity of the service accelerator's associated location. In variations, the service accelerator **152** for such default locations may also be constrained by one or more time windows (e.g., indicating a start time and an end time for displaying the accelerator **152**) based on the routine behavior of the user **174**.

As provided herein, the user information **147** can comprise historical data corresponding to the user's **174** utilization of the on-demand network-based service. In analyzing the user information **147**, the accelerator engine **150** can identify certain patterns with regard to the user **174** submitting service requests **171**. These patterns can include correlations between specific service locations, pick-up locations, destinations, time of day, day of the week, day of the year (e.g., a holiday, anniversary, birthday, etc.), season (e.g., a sports season), and the like. In some examples, if an identified pattern in the user information **147** meets a threshold level confidence or likelihood, then the accelerator engine **150** can generate a new service accelerator **152** based on the identified pattern. This new service accelerator **152** can be associated with certain accelerator triggers **179** corresponding to a set of criteria that causes the accelerator **152** to be displayed on the requester device **170**.

As an example, the requester profile **144** for a particular user **174** may indicate that the user **174** typically utilizes an on-demand transportation service in which the user **174** visits a certain location at a certain time (e.g., a specific day of the week, time of day, and/or day of the year). The location may be a place of work, a city center, a train station, airport, a sporting venue, a restaurant, a church, a conference center, a hotel, a park, a cemetery, etc. The pattern identified by the accelerator engine **150** can correspond to the user **174** utilizing the on-demand transport service to visit the location on multiple occasions (e.g., an amusement park on Father's Day). In some examples, the accelerator engine **150** can calculate a probability that the user **174** will utilize the transportation service at a future time, and if the probability exceeds a certain threshold (e.g., 45%), the accelerator engine **150** can push a service accelerator **152** for the location at the appropriate time. Additionally or alternatively, the service accelerator **152** for the location can be presented on the requester device **170** for a certain period of time (e.g., three hours), and can expire or terminate from the display screen once the time period ends.

In addition, the network computer system **100** can include an application synchronizer **160**, which can synchronize with the requester device **170** to access third party applications on the requester device **170**, such as a calendar application, a travel application, e-mail, a contacts list, and the like (collectively "third party app data **162**"). According to certain examples, the accelerator engine **150** can receive third party app data **162** from the application synchronizer **160** in order to determine whether a service accelerator **152** would be suitable or would otherwise assist the user **174**. The accelerator engine **150** can further combine the third party app data **162** with the user information **147** to determine whether to generate a service accelerator **152**. Comparisons between third party app data **162** and user information **147** can result in a wide range of correlations with varying complexity. For example, a calendar entry on a requester device **170** of a first user may indicate a meeting at a specified location with a second user in the contacts list of the first user. The accelerator engine **150** can identify the second user, and generate a service accelerator **152** for the second user at the specified location, based on the meeting indicated in the calendar entry of the first user.

In further examples, the third party application data **162** may indicate that the user **174** is scheduled for a flight from a local airport. Based on the user's travel schedule, the accelerator engine **150** can generate a destination accelerator **152** for the local airport for display on the requester's device **170** at a certain time prior to the scheduled flight departure (e.g., three hours prior). Still further, if the user **174** is located in a new place—such as a new city in which the user **174** has not previously utilized the on-demand network-based service—the accelerator engine **150** can identify certain interest locations for the user **174** and generate destination accelerators **152** for such interest locations. These locations may be general points of interest within the new place (e.g., popular tourist destinations), or can be customized for the user **174** based on interest patterns identified in the third party application data **162** and/or the user information **147**. In one example, the user **174** may show an interest in highly-regarded restaurants, or have a favorite type of food. The accelerator engine **150** may identify a restaurant that matches the user's **174** interests, and generate a service accelerator **152** for that restaurant for display on the requester's device **170**.

Thus, a virtually endless number of data combinations and correlations can result in the accelerator engine **150** generating a service accelerator **152** for a particular user **174**. Once criteria are determined for a service accelerator **152**, then the accelerator engine **150** can monitor current user conditions (e.g., the user's current location, a time of day, a schedule, etc.) for acceleration triggers **177**. When a set of acceleration triggers **177** matches the set of criteria for a particular service accelerator **152**, the accelerator engine **150** can generate the accelerator **152** to be displayed on the requester's device **170**.

As provided herein, display of the service accelerators **152** on the requester devices **170** can correspond to a selectable feature on a home screen of the executing service application **175**. In certain implementations, if the service application **175** is not currently executing, then the service accelerator **152** can be queued for display once the service application **175** is launched. Additionally, certain service accelerators **152** may be associated with a timer so that they expire and are automatically removed from the display. For example, the accelerator engine **150** can configure a work service accelerator **152** to expire at a local time of 10:00 am. As such, certain service accelerators **152** may be generated by the accelerator engine **150**, but may expire without actually being displayed on the requester device **170**.

In some variations, the accelerator engine **150** can generate and transmit a push notification to the user device **170** indicating that a service accelerator **152** has been triggered and is available for usage. This push notification can be provided to the requester device **170** regardless of which application, if any, is currently executing on the user device **170**. The push notification can comprise any combination of a displayed indicator, a unique sound (e.g., a chime unique to the service application **175**), and/or a haptic output on the requester device **170**. This can enable the user **174** to identify the service accelerator **152** without launching the service application **175**, and determine whether the user **174** wishes to utilize the service accelerator **152**.

According to examples described herein, the user **174** can provide an accelerator input **176** on a particular service accelerator **152** displayed on the requester device **170**. In many examples, the accelerator input **176** can comprise a touch selection on the displayed service accelerator **152**, or can comprise a voice command or other predetermined gesture (e.g., a swipe gesture on the accelerator **152**). In response to the accelerator input **176**, the selection engine **130** can identify the destination corresponding to the selected accelerator **152**, and automatically request a service provider to fulfill the service request corresponding to the service accelerator **152**.

The selection engine **130** can process a service accelerator input **176** from a requester device **170** by initially determining a service location for the requesting user **174**. In some aspects, the selection engine **130** can determine a user location **178** corresponding to the requesting user's **174** current location (e.g., via GPS resources of the requester device **170**). The selection engine **130** may then independently configure a service location for the requesting user **174** based on the current location. For example, the selection engine **130** can identify the user's current address as the service location, or a nearest convenient street location or address that can function as a rendezvous point between the requesting user **174** and an optimal service provider **189** to fulfill the service request **171**.

Alternatively, in response to the service accelerator input **176**, the content engine **120** can cause a confirmation screen to be generated on the user app **175**, which can query the requesting user **174** to confirm the service location. In either case, based on the user's **174** selection of the service accelerator **152**, the selection engine **130** can identify a candidate set of service providers **184** (e.g., utilizing a geo-fence centered on the user's **174** current location), and select an optimal service provider **189** to service the ride. Thereafter, the selection engine **130** can generate and transmit a service invitation **132** to the optimal service provider **189** via the service provider app **185** executing on the service provider device **180**. In response to an acceptance **181** of the service invitation **132**, the selection engine **130** can transmit a confirmation **134** to the requesting user **174** indicating that a service provider **189** is en route to the service location.

In variations, different input types on a service accelerator **152** can cause different responses. For example, in executing the service application **175** the requester device **170** can detect swipe gestures, tap and hold gestures (long press), and/or single tap gestures being performed by the user **174** on a service accelerator **152**. In some aspects, a tap and hold gesture can cause a service request **171** to be sent automatically to the network computer system **100**. In such aspects, while the service request **171** is being processed by the selection engine **130**, the service application **175** can display a processing screen (e.g., including a cancelation feature) indicating that a service provider is being matched before displaying an en route screen. Additionally or alternatively, a normal tap gesture (short press) on a service accelerator **152** can cause the service application **175** to display a confirmation screen with the information corresponding to the service accelerator **152** (e.g., the service location and service type) prepopulated.

FIG. **2** is a block diagram illustrating an example service accelerator system utilized in connection with a network computer system, according to examples described herein. The service accelerator system **200** shown and described with respect to FIG. **2** can comprise one or more logical blocks or components as shown and described with respect to FIG. **1**. For example, the service accelerator system **200** can be connected to network-based service subsystems **280**, which can comprise certain components of FIG. **1**, such as the application interface **125**, selection engine **130**, and content engine **120** of the network computer system **100**. Referring to FIG. **2**, the service accelerator system **200** can include a database **240** that stores requester profiles **242** of users of the on-demand network-based service in a given region. In various examples, the requester profiles **242** can include all or the most recent services **244** consumed by the user (e.g., rides consumed through an on-demand transport service), routine routes **246** for the user, and/or preconfigured service accelerators **248** for certain on-demand service types (e.g., default destination accelerators for the user, such as for home and/or work).

The service accelerator system **200** can include a profile manager **230** that can receive service data **218** for each user, and utilize the service data **218** to generate profile updates **232** for a given user's profile **242**. The service data **218** can include data indicating each service consumed by a given user, and details of the service, such as a service location, a transportation destination, any stops made, a time of day for the service, a day of the week (e.g., weekday versus weekend), weather information, traffic data, and the like. Accordingly, the profile manager **232** can update the consumed services **244** in the user's profile **242** based on the service data **218** received from the network-based service subsystems **280**.

In some examples, the service accelerator system **200** can further include an application synchronization module **220** that can identify when a particular requester device **295** launches the service application **297**. This detection can occur via an application launch trigger **219** from the service application **297** on the requester device **295**. The application launch trigger **219** can cause the application synchronization module **220** to generate a synchronization request **227** to access third party data and other user data on the requester device **295**. In one example, the synchronization request **227** can enable the user to accept or decline the service accelerator system **200** in its request to accept the user data. If the user accepts, then the application synchronization module **220** can parse through various data sources, such as a contacts list, appointment calendar, travel plans, social media information, and the like. The application synchronization module **220** can pull the user's contacts **223**, any upcoming appointments **221**, and other third party application data **211** from the requester device **295** to enable the profile manager **230** to update the user's requester profile **242** in the database **240**.

In certain examples, data items including destination, time, and date information (e.g., appointments **221** or a travel itinerary) can cause the profile manager **230** to generate a preconfigured accelerator **248** in the requester profile **242**. The preconfigured accelerator **248** can include preset accelerator triggers **216** that can cause an accelerator generator **250** to generate a service accelerator **259** for display on the requester device **295**. Other information, such as the third party application data **211** and the contacts **223** can aid in enabling the accelerator generator **250** to determine the manner in which new service accelerators **259** may be generated.

The service accelerator system **200** can include a timer/calendar **265** that provides time and date information **267** to the accelerator generator **250**, and a map engine **235** that can provide map data **237** to the accelerator generator **250**. According to examples described herein, the accelerator generator **250** can receive user location data **213** from a respective requester device **295** when the service application **297** is executing thereon. In some aspects, the user location **213** can act as an accelerator trigger **216** when the user enters a certain geo-fenced area associated with a service accelerator **259**, or leaves a location associated with a service accelerator **259** (e.g., the user's home). Furthermore, the accelerator generator **250** can be programmed to or otherwise execute logic that causes the accelerator generator **250** to create a service accelerator **259** by imparting a set of criteria based on information specific to the user, or even information not specific to the user (e.g., general popular destinations, scheduled events, etc.). When satisfied by observed accelerator triggers **216**, these criteria can cause the accelerator generator **250** to cause the service accelerator **259** to be displayed on the requester device **295**. In certain implementations, the accelerator generator **250** can perform lookups **256** in the requester profiles **242** or otherwise receive profile data **249** from the requester profiles **242** in order to configure the set of criteria for a service accelerator **259**.

The accelerator generator **267** can monitor the time and date information **267**, the user locations **213** in the map data **237**, and other accelerator triggers **216**. Once the set of criteria for a particular service accelerator **259** are triggered by the observed accelerator triggers **216**, the accelerator generator **250** can generate the service accelerator **259** for display on the requester device **295** (e.g., a home screen on the service application **297**). As provided herein, the user can select the service accelerator **259** to request an on-demand service automatically. In one aspect, the single input selection of the service accelerator **259** causes the network-based service subsystems **280** to automatically request a specified ride service type for the user, and a service location based on the current user location data **213** from the requester device **295**. Thus, the user's service type preferences can also be indicated in the user's profile **242**. As described herein, for on-demand transport, the ride service type can comprise any of a ride-pooling service, a basic ride share service, a luxury vehicle service, a van or large vehicle service, a professional driver service (e.g., where the service provider is certified), a self-driving vehicle transport service, and the like.

Furthermore, the service type preference of the user can be context specific (e.g., time and/or date specific, or location specific). For example, the service data **218** may indicate that the user prefers the basic ride-share service during the day and a luxury transport service at night. Accordingly, the default selection of a ride service type (e.g., by the selection engine **130** of FIG. **1**) may be based on the current context of the user's state. Alternatively, the user may be requested to confirm information such as an estimated or upfront price for the service and/or the selected service type.

In various aspects, the accelerator generator **250** can continue to monitor accelerator triggers **216** to determine when to remove the service accelerator **259** from being displayed. As a basic example, if the user drives or walks home, the accelerator generator **250** can cause the home destination accelerator **259** to be removed from the home screen of the service application **297**. In more multifaceted examples, a service accelerator **259** having multiple requirements may be removed when one of the requirements ceases to apply. This requirement can be a time expiration, a location expiration, situational change triggers (e.g., weather, traffic, event schedules, etc.), and any combination of the foregoing.

Methodology

FIG. **3**A is a flow chart describing an example method of generating a service accelerator, according to examples described herein. In the below discussion of the FIG. **3**A, reference may be made to reference characters representing like features as shown and described with respect to FIGS. **1** and **2**. Furthermore, the processes described with respect to FIG. **3**A may be performed by an example network computer system **100** implementing a service accelerator system **200** as shown and described with respect to FIGS. **1** and **2**. Referring to FIG. **3**A, the network computer system **100** can manage an on-demand network-based service for a given region (**300**). For each user of the one-demand network-based service, the network computer system **100** can create a number of service accelerators **152** (**305**). As described herein, the service accelerator **152** can be displayed on the requester device **170** of the user (e.g., on a home screen of an executing service application **175**), and can be selectable to cause the network computer system **100** to automatically request a service for the user. Furthermore, the service accelerators **152** may be specific to the user, and can be attributed to a specified on-demand network-based service, such as a bill payment service, a repair service, delivery service, or transportation service. For on demand transportation services, the service accelerator **152** can further include common locations of the user, such as a work location, a home location, a gym location, a store location, and the like.

In some examples, the network computer system **100** can receive an app launch indication from a requester device **170**, indicating that the user **174** has launched the service application **175** (**310**). Based on the app launch indication, the network computer system **100** can determine the current location **173** of the user **174** (**315**). Based on the current location **173** of the user **174**, the network computer system **100** can determine whether to push one or more service accelerators **152** to be displayed on the requester device **170** of the user (**320**). For example, if the user is located at home, the network computer system **100** will not cause a home destination accelerator **152** for a transportation service to be displayed. However, one or more other default accelerators **152** may be displayed, such as a "work" destination accelerator, or a "store" destination accelerator.

Accordingly, if the current location **173** of the user **174** corresponds to a trigger **177** to display one or more service accelerators **152** (**322**), the network computer system **100** can generate and present the service accelerator(s) **152** on a user interface of the service application **175**. However, if the current location **173** of the user **174** does not correlate with a trigger **177** for any service accelerators **152** associated with the user **174** (**324**), then the network computer system **100** can continue to monitor the user's dynamic location (**345**).

For service accelerators **152** displayed on the user interface of the service application **175**, the network computer system **100** can detect a user input **176** on a particular service accelerator **152** (**330**). In some examples, the user input **176** can trigger the network computer system **100** to generate one or more confirmation screen(s) to be displayed on the requester device **170** to confirm the default service type associated with the selected service accelerator **152**, the estimated or actual price, and/or enable the user **174** to modify the service request **171** (**350**). Additionally or alternatively, the network computer system **100** can automatically identify a service location in proximity to or at the user's current location **173**, and select an optimal service provider **189** to facilitate the service request **171** corresponding to the selected service accelerator **152** (**335**). Once the optimal service provider **189** is selected, the network computer system **100** can transmit a service invitation **132** to the computing device **180** of the optimal service provider **189** via an executing service provider application **185** to fulfill the service request **152** (**340**).

FIG. **3**B is a flow chart describing an example method of pivoting a user interface to enable input of one or more service locations (e.g., a pick-up location and a destination location for an on-demand transportation service), according to examples described herein. The network computer system **100** can provide a user interface for a service application **175** to be displayed on a requester device **170** (**355**). For example, the user interface can include a screen to enable the user **174** to input a service location in order to submit a service request **171** for the on-demand network-based service. As provided herein, this user interface screen can dynamically pivot between two configurations, which can affect the manner in which the user **174** inputs a service location. Thus, a first pivot configuration can include a default map screen with a location pin in which the user can scroll the map content in order to set the location pin as the service location (**357**). A second pivot configuration may be displayed in response to a specified user input on the default map screen, such as a selection input on an address input box that is persistently displayed in the first pivot configuration. The second pivot configuration can enable the user to manually input an address for the service location, and can include a keyboard to enable the input (**359**).

In the first pivot configuration, the requester device **170** can receive a scroll input on the map content to input the service location (**360**). Based on the scroll input, the requester device **170** can cause the map content to scroll accordingly, enabling the user **174** to input the location pin at a certain desired location on the map (**365**). In some aspects, once the location pin is set, the requester device **170** can find a nearest street location (**367**), and define the location as the service location (**369**). Accordingly, the user **174** is enabled to set a service location by scrolling the map content displayed on the requester device **170**.

In certain implementations, the first pivot configuration can also include a persistent input feature (e.g., an input box) that enables the user to manually input an address or location for the service location. At any given time when the user interface is in the first pivot configuration, the requester device **170** can receive a user input on the service location input box (**370**). In response to the input, the requester device **170** can dynamically pivot the first configuration to the second configuration to display at least a keyboard and the input box (**375**). In further examples, the pivot can also cause one or more "address" or "places" panels to be displayed in the second configuration in order to further aid the user **174** in inputting the service location. Furthermore, the user input on the input box can disable map scrolling functions while causing the additional panels (e.g., keyboard, a predictive address panel, a favorite locations panel, etc.) to overlay or replace the map content. As provided herein, the service location can comprise a location at which a selected service provider is to rendezvous with the requesting user, or where the service provider is to perform the requested service. In the context of on-demand transport services, the service location(s) can comprise a pick-up location, a destination location, or both.

This pivoting mechanism may be distinct from screen-loading, as the dynamic pivoting between the two configurations can occur on a single screen using cached or other stored or readily pulled content items to enable the pivoting configurations. Thus, while the user interface is in the second pivot configuration, the requester device **170** can receive a specified user gesture (e.g., a scroll input) (**380**), and in response to the gesture, pivot the user interface back to the first configuration by removing any additional panels and/or the keyboard from the user interface, and re-enabling map content scrolling (**385**). As discussed herein, when the user **174** inputs the location pin on a desired location, the requester device **170** can set the location pin on the scrollable map content as the service location or destination (**390**). However, at any given time prior to inputting the service location or destination, the requester device **170** can receive an input on the display input box to once again pivot the user interface to the second pivot configuration (**370**).

The sequencing of user interface screens can be provided in any number of suitable manners. In certain examples, once the user **174** enters the service location in either the first or the second pivot configuration, the service application **175** can enable the user to submit a service request **171**. For transport services, a new pivoting interface like the one discussed above, may be displayed to enable the user **174** to enter a desired destination location. In certain examples, the pivoting interface can be displayed if the user opts not to utilize a service accelerator **152** on a home screen of the service application **175**. In other examples, the pivoting interface can be displayed on the service application whenever the user wishes to input an address or location using the service application **175**, whether inputting a service location, a rendezvous point, a destination, or a place of interest. Further description of the pivoting interface is provided below in connection with FIG. **6**C.

FIG. **4** is a flow chart describing an example method of configuring a service accelerator and facilitating on-demand services, according to examples described herein. As discussed above, the network computer system **100** can manage an on-demand transportation service for a given region (**400**). In certain examples, the network computer system **100** can synchronize with a requester device **170** to receive third party application data **162** specific to the user **174** (**405**). This third party application data **162** can include contacts from a contacts application (e.g., a phonebook or a social media application) (**406**), the user's schedule from a calendar application (**407**), a travel itinerary or schedule from a travel application (**408**), and the like.

In various aspects, the network computer system **100** can manage requester profiles **144** and can configure service accelerators **152** for each of the users **174** (**410**). The requester profiles **144** can indicate common destinations for the user **174**—such as a home location, a work location, and other frequented locations by the user **174**—which the network computer system **100** can utilize to set a number of default accelerators **152**. Furthermore, the network computer system **100** can associate each service accelerator **152** with a set of requirements or criteria that, when satisfied, cause the network computer system to generate the service accelerator **152** on the requester device **170**. Accordingly, each service accelerator **152** may be associated with a particular destination location (**411**), a time of day and/or day of the week (**412**), and/or an on-demand service type (**413**).

According to examples, the network computer system **100** can identify accelerator triggers **179** based on the current state of the user **174** (**415**). For example, the network computer system **100** can receive data corresponding to the current location **173** of the user **174** (e.g., from location-based resources on the requester device **170**). The location **173** of the user **174** can enable the network computer system **100** to identify location triggers (**417**) that cause or contribute to causing a service accelerator **152** to be displayed on the requester device **170**. Further, the network computer system **100** can further identify time of day and/or day of the week triggers (**416**). Still further, the network computer system **100** can identify appointment triggers (**418**), in which time and location are specified (e.g., a meeting, a conference, a travel plan, etc.). When the requirements for a particular service accelerator **152** are met, the network computer system **100** can generate a service accelerator **152** for display via the service application **175** executing on the user's device (**420**).

In certain aspects, the network computer system **100** can detect a user input **176** on the displayed service accelerator **152** (**425**). In one implementation, the user input **176** can cause the network computer system **100** to generate one or more confirmation screens to confirm service details, such as price and service type (**430**). Alternatively, the user input **176** can trigger the network computer system **100** to identify a service location for the user (**435**). The service location can be based on a current location **173** of the user **174** (e.g. via GPS resources of the requester device **170**), or can be a predetermined location based on past utilization the on-demand network-based service (e.g., a convenient street corner as a pick-up location, or a location near the user's home) (**439**).

The network computer system **100** can then identify a set of service providers **184** within a location proximity or estimated time from the service location (**440**), and select an optimal service provider **189** to service the accelerator ride request (**445**). Thereafter, the network computer system **100** can transmit a service invitation **132** to the optimal service provider **189** to fulfill or otherwise complete the service request, which the service provider **189** can accept or decline (**450**). If the service provider **189** declines, then the network computer system **100** can identify another optimal service provider **189**, either from the candidate set of service providers **184**, or from a new set of service providers based on the service location. However, if the service provider accepts, then the network computer system **100** can receive an indication of the acceptance confirmation **181** from the optimal service provider **189** (**455**), and transmit a rendezvous confirmation to the requesting user **174** (**460**).

User Device

FIG. **5** is a block diagram illustrating an example user device executing a designated service application for an on-demand network service, as described herein. In many implementations, the user device **500** can comprise a mobile computing device, such as a smartphone, tablet computer, laptop computer, VR or AR headset device, and the like. As such, the user device **500** can include typical telephony features such as a microphone **545**, a camera **550**, and a communication interface **510** to communicate with external entities using any number of wireless communication protocols. In certain aspects, the user device **500** can store a designated application (e.g., a service app **532**) in a local memory **530**. In many aspects, the user device **500** further store information corresponding to a contacts list **534**, and calendar appointments **536** in the local memory **530**. In variations, the memory **530** can store additional applications executable by one or more processors **540** of the user device **500**, enabling access and interaction with one or more host servers over one or more networks **580**.

In response to a user input **518**, the service app **532** can be executed by a processor **540**, which can cause an app interface **542** to be generated on a display screen **520** of the user device **500**. The app interface **542** can enable the user to, for example, check current price levels and availability for the on-demand network-based service. In various implementations, for transport services, the app interface **542** can further enable the user to select from multiple service types, such as a carpooling service, a regular ride-sharing service, a professional ride service, a van or high-capacity vehicle transport service, a luxurious ride service, and the like.

The user can generate a service request **567** via user inputs **518** provided on the app interface **542**. For example, the user can select a service location, view the various service types and estimated pricing, and select a particular service type (e.g., for transportation to an inputted destination). As provided herein, the service application **532** can further enable a communication link with a network computer system **590** over the network **580**, such as the network computer system **100** as shown and described with respect to FIG. **1**. Furthermore, as discussed herein, the service application **532** can enable the network computer system **590** to cause service accelerators **528** to be displayed on the application interface **542**. The service accelerators **528** can be selected to automatically cause a service request to be generated and transmitted to the network computer system **590** (e.g., with preconfigured service locations and/or destinations).

The processor **540** can transmit the service requests **567** via a

communications interface **510** to the backend network computer system **590** over a network **580**. In response, the user device **500** can receive a confirmation **569** from the network computer system **590** indicating the selected service provider that will fulfill the service request **567** and rendezvous with the user at the service location. In various examples, the user device **500** can further include a GPS module **560**, which can provide location data **562** indicating the current location of the requesting user to the network computer system **590** to, for example, establish the service location and/or select an optimal service provider to fulfill the service request **567**.

User Interface Screenshots

FIGS. **6**A through **6**C illustrate example screenshots of a user interface on a user device, according to examples described herein. In the below description of FIGS. **6**A through **6**C, reference may be made to reference characters representing like features as shown and described with respect to FIG. **5**. Furthermore, in the context of FIGS. **6**A through **6**C, the on-demand network-based service can comprise an on-demand transportation service utilizing destination accelerators as the service accelerators. Thus, a service provider can comprise a driver, and service request can comprise a request for pick-up to cause the on-demand transportation service to select an optimal driver to rendezvous with the requesting user at a pick-up location, and transport the requesting user to an inputted destination.

Referring to FIG. **6**A, execution of the service application **532** on the user device **500** can cause an app interface **601** to be generated on the display screen **520**. In some aspects, the app interface **601** can comprise an initial home screen **600**, and can feature such elements as a destination input box **603**, a location feature **605** indicating the user's current location, and virtual representations of proximate available drivers **607**. According to examples provided, the home screen **600** can also include a number of destination accelerators **609** that are selectable to automatically submit a pick-up request without additional input or manual intervention by the user, or with minimal additional input (e.g., a confirmation selection on a subsequent screen).

FIG. **6**B illustrates an "en route" screen **610** generated once a pick-up request has been submitted and a servicing driver **615** has been selected. In certain aspects, the en route screen **610** can be generated as a subsequent screen in response to a user selection of a service accelerator in the form of a destination accelerator **609** on the home screen. Thus, a single input on the destination accelerator **609** can cause the screen shown in FIG. **6**B to be generated. The en route screen **610** can include a pick-up location **613** as well as the user's current location, and estimated time of arrival information **611** for the servicing driver **615**. In many aspects, the en route screen **610** can further include driver information **617** such as the servicing driver's **615** name, vehicle type, and license plate number.

FIG. **6**C shows a screen sequence chain indicating a service confirmation screen **620**, a location input screen **630**, and a pivot screen **640** having a first pivot configuration **650** and a second pivot configuration **660**. In some examples, the service configuration screen **620** can be presented as a confirmation screen in response to a user selection of a destination accelerator **609**, as shown in FIG. **6**A. In other examples, the service configuration screen **620** can be presented in response to a user setting a pick-up location pin **649**, and submitting a pick-up request. In such examples, the user may then configure the destination manually by inputting a destination selection **621**, which can cause the location input screen **630** or the pivot screen **640** to be displayed. As provided herein, selection of a destination accelerator **609** (shown in FIG. **6**A) can bypass a number of these additional steps.

The service confirmation screen **620** can display a plurality of service types **624** and price information **622** corresponding to each service type. As shown on the service screen **620**, the service types **624** include a combination of walking and ride-pooling services, a ride-pooling service, and a standard ride-sharing service. As described herein, it is contemplated that destination accelerators displayed on the home screen **600**, and associated with a particular ride service type, can include the price information **622** shown in FIG. **6**C—which can comprise an estimated price or an upfront, guaranteed price for a requested ride. For destination accelerator implementations, a default service **626** may be highlighted and the user can confirm the ride request by selecting a request trigger **628**, thereby triggering the en route screen **610** of FIG. **6**B. The price information **624** can comprise an estimated cost of the trip or an upfront price that may be guaranteed for the trip. In addition, the service confirmation screen **620** can enable the user to view proposed route information **627** between a current location **625** and the inputted or preconfigured destination. Furthermore, in some examples, the service confirmation screen can provide estimated time of arrival information **629** for the resultant ride, which can be based on current traffic conditions.

In certain implementations, the user can modify the pick-up location and/or destination location by selecting either the current location feature **625** or the destination selection feature **621**. In either case, one of the location input screen **630** or the pivot screen **640** may be displayed. In one example, the pivot screen **640** is displayed only for pick-up location configuration in response to a user selection of the current location feature **625**. Additionally or alternatively, the pivot screen **640** is displayed upon a user selection of the destination selection feature **621** to enable the user to configure the destination location. In variations, the location input screen **630** is displayed to enable the user to manually input—using a displayed keyboard **637**—the pick-up location via a pick-up location input box **633**, and/or the destination location via a destination input box **635**. In such variations, the interface panels and be generated to overlay underlying map content **639**.

For pivot screen **640** examples, selection of the current location feature **625** or the destination selection feature **621** can cause a first pivot configuration **650** to be generated. The first pivot configuration **650** can comprise scrollable map content **648** and a location pin **649** to enable the user to set a location—whether a destination or a pick-up location. In the example shown, the first configuration **650** can also include an input box **641** that enables the user to provide a user input **644** on either the pick-up location box or the destination box. Throughout the description of FIGS. **6**A through **6**C, "input boxes" are shown as selectable to enable manual input. However, it is contemplated that any selectable feature, such as an icon or map indicator, can be selected to enable manual input (e.g., of a location), or may be inputted via voice command. As provided herein, the user input **644** on the input box **641** can trigger the second pivot configuration **660** to be generated on the pivot screen **640**, and can cause the scrollable nature of the map content **648** to be disabled.

Thus, the pivot screen **640** can comprise a single screen having pre-cached or otherwise pre-configured content, or readily accessed content without requiring significant content loading, that can be dynamically generated in response to the user input **644** on the input box **641**. In this example, the pre-configured content can comprise a number of panels and a keyboard that can assist the user in inputting a location (e.g., the pick-up location). In one example, the assistive panels can include a "favorites" panel indicating common locations for the user. Additionally or alternatively, the assistive panels can include a "suggestions" panel that can comprise locations relating to predictive text in the input box **641**, popular locations within proximity to the user, or locations predictive of the user's personal interests. At any given time while the second pivot configuration **660** is displayed, the user can provide a predetermined input **646** to cause the pivot screen **640** to return to the first pivot configuration **650**. In one example, the predetermined input **646** can comprise a swipe or scroll input anywhere on the display. Processing resources, via execution of the rider application on the rider device, can translate the swipe or scroll input—while the user interface displays the pivot screen **640** in the second pivot configuration **660**—into a command to dismiss the assistive panels and keyboard, re-enable the scrollable nature of the map content **648**, and generate the pivot screen **640** in the first pivot configuration **650**.

Upon input of the destination and/or pick-up location, the rider application can return to the service confirmation screen **620** to enable the user to submit the ride request. Furthermore, once the pick-up location, destination, and ride service type are configured, the user may select the request trigger **628**. In response to the request trigger **628**, the backend transport facilitation system **100** (shown and described with respect to FIG. **1**), can select a driver to service the request, and cause the en route screen **610** (of FIG. **6**B) to be presented on the display of the rider device.

The above description of the screen sequence chain of FIG. **6**C is illustrative of various example implementations of user interactions on the app interface **601** in order to make a ride request. However, as described herein, the use of preconfigured destination accelerators **609** can bypass a number of steps, such as the steps of inputting the destination, inputting the pick-up location, inputting the ride service type, and even inputting the ride request on the request trigger **628**. Thus, it is contemplated that one or more of the foregoing steps may be circumvented through user selection of a destination accelerator **609**. Consequently, it is also contemplated that all such input steps may be circumvented through user selection of a destination accelerator **609**—in which the app interface **601** transitions directly from the home screen **600** of FIG. **6**A, to the en route screen **610** of FIG. **6**B.

Service Provider Device

FIG. **7** is a block diagram illustrating an example service provider device executing a designated service provider application for an on-demand network service, as described herein. In many implementations, the service provider device **700** can comprise a mobile computing device, such as a smartphone, tablet computer, laptop computer, VR or AR headset device, and the like. As such, the service provider device **700** can include typical telephony features such as a microphone **745**, a camera **750**, and a communication interface **710** to communicate with external entities using any number of wireless communication protocols. The service provider device **700** can store a designated application (e.g., a service provider app **732**) in a local memory **730**. In response to a user input **718**, the service provider app **732** can be executed by a processor **740**, which can cause an app interface **742** to be generated on a display screen **720** of the service provider device **700**. The app interface **742** can enable the service provider to, for example, accept or reject service invitations **792** in order to fulfill service requests throughout a given region.

In various examples, the service provider device **700** can include a GPS module **760**, which can provide location data **762** indicating the current location of the service provider to the network computer system **790** over a network **780**. Thus, the network computer system **790** can utilize the current location **762** of the service provider to determine whether the service provider is optimally located to fulfill a particular service request. If the service provider is optimal to fulfill the service request, the network computer system **790** can transmit a service invitation **792** to the service provider device **700** over the network **780**. The service invitation **792** can be displayed on the app interface **742**, and can be accepted or declined by the service provider. If the service provider accepts the invitation **792**, then the service provider can provide a user input **718** on the displayed app interface **742** to provide a confirmation **722** to the network computer system **790** indicating that the service provider will rendezvous with the requesting user at the service location to fulfill the service request.

Hardware Diagram

FIG. **8** is a block diagram that illustrates a computer system upon which examples described herein may be implemented. A computer system **800** can be implemented on, for example, a server or combination of servers. For example, the computer system **800** may be implemented as part of a network-based service for providing transportation services. In the context of FIG. **1**, the network computer system **800** may be implemented using a computer system **800** such as described by FIG. **8**. The network computer system **100** may also be implemented using a combination of multiple computer systems as described in connection with FIG. **8**.

In one implementation, the computer system **800** includes processing resources **810**, a main memory **820**, a read-only memory (ROM) **830**, a storage device **840**, and a communication interface **850**. The computer system **800** includes at least one processor **810** for processing information stored in the main memory **820**, such as provided by a random access memory (RAM) or other dynamic storage device, for storing information and instructions which are executable by the processor **810**. The main memory **820** also may be used for storing temporary variables or other intermediate information during execution of instructions to be executed by the processor **810**. The computer system **800** may also include the ROM **830** or other static storage device for storing static information and instructions for the processor **810**. A storage device **840**, such as a magnetic disk or optical disk, is provided for storing information and instructions.

The communication interface **850** enables the computer system **800** to communicate with one or more networks **880** (e.g., cellular network) through use of the network link (wireless or wired). Using the network link, the computer system **800** can communicate with one or more computing devices, one or more servers, and/or one or more self-driving vehicles. In accordance with examples, the computer system **800** receives service requests **882** from mobile computing devices of individual users. The executable instructions stored in the memory **830** can include selection instructions **822**, which the processor **810** executes to select an optimal service provider to fulfill or complete the service request **882**. In doing so, the computer system can receive service provider locations **884** of service providers operating throughout the given region, and the processor can execute the selection instructions **822** to select an optimal service provider from a set of available service providers, and transmit a service invitation **852** to enable the service provider to accept or decline the requested service.

The executable instructions stored in the memory **820** can also include accelerator instructions **824**, which enable the computer system **800** to access user profiles **826** and other user information in order to generate service accelerators **854** for display on the user devices. As described extensively throughout, these service accelerators **854** can be selectable to automatically generate service requests **882**, bypassing a number of superfluous steps when such accelerators **854** are utilized.

By way of example, the instructions and data stored in the memory **820** can be executed by the processor **810** to implement an example network computer system **100** of FIG. **1**. In performing the operations, the processor **810** can receive service requests **882** (e.g., via manual input or service accelerators **854**) and service provider locations **884**, and submit service invitations **852** to facilitate the servicing of the requests **882**.

The processor **810** is configured with software and/or other logic to perform one or more processes, steps and other functions described with implementations, such as described by FIGS. **1**-**7**, and elsewhere in the present application.

Examples described herein are related to the use of the computer system **800** for implementing the techniques described herein. According to one example, those techniques are performed by the computer system **800** in response to the processor **810** executing one or more sequences of one or more instructions contained in the main memory **820**. Such instructions may be read into the main memory **820** from another machine-readable medium, such as the storage device **840**. Execution of the sequences of instructions contained in the main memory **820** causes the processor **810** to perform the process steps described herein. In alternative implementations, hard-wired circuitry may be used in place of or in combination with software instructions to implement examples described herein. Thus, the examples described are not limited to any specific combination of hardware circuitry and software.

It is contemplated for examples described herein to extend to individual elements and concepts described herein, independently of other concepts, ideas or systems, as well as for examples to include combinations of elements recited anywhere in this application. Although examples are described in detail herein with reference to the accompanying drawings, it is to be understood that the concepts are not limited to those precise examples. As such, many modifications and variations will be apparent to practitioners skilled in this art. Accordingly, it is intended that the scope of the concepts be defined by the following claims and their equivalents. Furthermore, it is contemplated that a particular feature described either individually or as part of an example can be combined with other individually described features, or parts of other examples, even if the other features and examples make no mentioned of the particular feature. Thus, the absence of describing combinations should not preclude claiming rights to such combinations.

What is claimed is:

1. A computing system comprising:

one or more processors;

a memory to store a set of instructions;

wherein the one or more processors execute the set of instructions to perform operations that include:

storing, in a database, historical data for a plurality of users, the historical data for each user relating to a utilization, by that user, of a network service for receiving transport, the historical data including destination locations;

associating each user of the plurality of users with a corresponding set of destination locations based on the historical data for that user;

communicating, over one or more networks, with a computing device of each user of the plurality of users to determine a current location of the computing device, as determined by a GPS resource of the computing device, in connection with the computing device launching a service application for enabling the user to utilize the network service;

for each user of the plurality of users, (i) determining, based on the current location of the computing device of the user, whether or not to display, for each destination location of the corresponding set of destination locations, a corresponding destination accelerator feature, and (ii) based on the determination, transmitting data to the computing device of the user upon a corresponding service application being launched on the computing device of the user, without manual input from the user, the data causing the corresponding destination accelerator feature for each of one or more destination locations of the corresponding set of destination locations to be displayed on the computing device of the user, each displayed destination accelerator feature being selectable to generate a transport request for transporting the user to the destination location that corresponds to the selected destination accelerator feature, without the user providing additional input to identify the destination location that corresponds to the selected destination accelerator feature;

wherein for at least a first user of the plurality of users, the transmitted data causes the corresponding service application of the computing device of the first user to display at least the destination accelerator feature for a first destination location of the corresponding set of destination locations for the first user, without displaying the destination location feature for at least a second destination location of the corresponding set of destination locations;

based on the first user selecting the destination accelerator feature for the first destination location, receiving, over the one or more networks from the computing device of the first user, a service request to transport the first user to the first destination location; and

processing the service request by selecting, from a plurality of candidate service providers, a service provider to fulfill the service request based, at least in part, on a current location of the service provider, as determined by a GPS resource of a computing device of the service provider.

2. The computing system of claim **1**, wherein determining whether or not to display each destination accelerator feature of the corresponding set of destination locations is further based at least in part on contextual information.

3. The computing system of claim **2**, wherein the contextual information includes a time of day, or a day of a week.

4. The computing system of claim **1**, wherein each displayed destination accelerator feature is selectable to generate the transport request for a selected type of service of multiple types of services, based on the historical data associated with the first user.

5. The computing system of claim **4**, wherein the multiple types of services includes at least one of a luxury vehicle service, a carpool service, a high capacity vehicle service, a professional driver service, or a self-driving vehicle service.

6. The computing system of claim **1**, wherein for at least the first user of the plurality of users, the transmitted data causes the computing device of the first user to display the destination accelerator feature the first destination location for a time window, after which the destination accelerator feature for the first destination location expires and is removed from a screen of the corresponding service application.

7. The computing system of claim **1**, wherein for at least the first user of the plurality of users, the destination accelerator feature of the first destination is associated with one or more conditions, and wherein the operations further comprise communicating with the computing device of the first user to determine when the associated condition of the destination accelerator feature ceases to apply, and removing the destination accelerator feature from a screen of the corresponding service application upon making the determination.

8. A non-transitory computer-readable medium that stores instructions, which when executed by one or more processors of a computing system, cause the computing system to perform operations that include:

storing, in a database, historical data for a plurality of users, the historical data for each user relating to a utilization, by that user, of a network service for receiving transport, the historical data including destination locations;

associating each user of the plurality of users with a corresponding set of destination locations based on the historical data for that user;

communicating, over one or more networks, with a computing device of each user of the plurality of users to determine a current location of the computing device, as determined by a GPS resource of the computing device, in connection with the computing device launching a service application for enabling the user to utilize the network service;

for each user of the plurality of users, (i) determining, based on the current location of the computing device of the user, whether or not to display, for each destination location of the corresponding set of destination locations, a corresponding destination accelerator, and (ii) based on the determination, transmitting data to the computing device of the user upon a corresponding service application being launched on the computing device of the user, without manual input from the user, the data causing the corresponding destination accelerator feature for each of one or more destination locations of the corresponding set of destination locations to be displayed on the computing device of the user, each displayed destination accelerator feature being selectable to generate a transport request for transporting the user to the destination location that corresponds to the selected destination accelerator feature, without the user providing additional input to identify the destination location that corresponds to the selected destination accelerator feature;

wherein for at least a first user of the plurality of users, the transmitted data causes the corresponding service application of the computing device of the first user to display at least the destination accelerator feature for a first destination location of the corresponding set of destination locations for the first user, without displaying the destination location feature for at least a second destination location of the corresponding set of destination locations;

based on the first user selecting the destination accelerator feature for the first destination location, receiving, over the one or more networks from the computing device of the first user, a service request to transport the first user to the first destination location; and

processing the service request by selecting, from a plurality of candidate service providers, a service provider to fulfill the service request based, at least in part, on a current location of the service provider, as determined by a GPS resource of a computing device of the service provider.

9. The non-transitory computer-readable medium of claim **8**, wherein determining whether or not to display each destination accelerator feature of the corresponding set of destination locations is further based at least in part on contextual information.

10. The non-transitory computer-readable medium of claim **9**, wherein the contextual information includes a time of day, or a day of a week.

11. The non-transitory computer-readable medium of claim **8**, wherein each displayed destination accelerator feature is selectable to generate the transport request for a selected type of service of multiple types of services, based on the historical data associated with the first user.

12. The non-transitory computer-readable medium of claim **11**, wherein the multiple types of services includes at least one of a luxury vehicle service, a carpool service, a high capacity vehicle service, a professional driver service, or a self-driving vehicle service.

13. The non-transitory computer-readable medium of claim **8**, wherein for at least the first user of the plurality of users, the transmitted data causes the computing device of the first user to display the destination accelerator feature the first destination location for a time window, after which the destination accelerator feature for the first destination location expires and is removed from a screen of the corresponding service application.

14. The non-transitory computer-readable medium of claim **8**, wherein for at least the first user of the plurality of users, the destination accelerator feature of the first destination is associated with one or more conditions, and wherein the operations further comprise communicating with the computing device of the first user to determine when the associated condition of the destination accelerator feature ceases to apply, and removing the destination accelerator feature from a screen of the corresponding service application upon making the determination.

15. A method for providing a network service, the method being implemented by one or more processors of a computing system and comprising:

storing, in a database, historical data for a plurality of users, the historical data for each user relating to a utilization, by that user, of a network service for receiving transport, the historical data including destination locations;

associating each user of the plurality of users with a corresponding set of destination locations based on the historical data for that user;

communicating, over one or more networks, with a computing device of each user of the plurality of users to determine a current location of the computing device, as determined by a GPS resource of the computing device. in connection with the computing device launching a service application for enabling the user to utilize the network service;

for each user of the plurality of users, (i) determining, based on the current location of the computing device of the user, whether or not to display, for each destination location of the corresponding set of destination locations, a corresponding destination accelerator feature, and (ii) based on the determination, transmitting data to the computing device of the user upon a corresponding service application being launched on the computing device of the user, without manual input from the user, the data causing the corresponding destination accelerator feature for each of one or more destination locations of the corresponding set of destination locations to be displayed on the computing device of the user, each displayed destination accelerator feature being selectable to generate a transport request for transporting the user to the destination location that corresponds to the selected destination accelerator feature, without the user providing additional input to identify the destination location that corresponds to the selected destination accelerator feature;

wherein for at least a first user of the plurality of users, the transmitted data causes the corresponding service application of the computing device of the first user to display at least the destination accelerator feature for a first destination location of the corresponding set of destination locations for the first user, without displaying the destination location feature for at least a second destination location of the corresponding set of destination locations;

based on the first user selecting the destination accelerator feature for the first destination location, receiving, over the one or more networks from the computing device of the first user, a service request to transport the first user to the first destination location; and

processing the service request by selecting, from a plurality of candidate service providers, a service provider to fulfill the service request based, at least in part, on a current location of the service provider, as determined by a GPS resource of a computing device of the service provider.

16. The method of claim **15**, wherein determining whether or not to display each destination accelerator feature of the corresponding set of destination locations is further based at least in part on contextual information.

17. The method of claim **16**, wherein the contextual information includes a time of day, or a day of a week.

18. The method of claim **15**, wherein each displayed destination accelerator feature is selectable to generate the transport request for a selected type of service of multiple types of services, based on the historical data associated with the first user.

19. The method of claim **18**, wherein the multiple types of services includes at least one of a luxury vehicle service, a carpool service, a high capacity vehicle service, a professional driver service, or a self-driving vehicle service.

20. The method of claim **15**, wherein for at least the first user of the plurality of users, the transmitted data causes the computing device of the first user to display the destination accelerator feature the first destination location for a time window, after which the destination accelerator feature for the first destination location expires and is removed from a screen of the corresponding service application.

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